Validation protocol reverses machine-learning model rankings in tokamak energy-confinement scaling

Liam Salcedo*

6 September 2026

Code, results, and reproducibility materials: `https://github.com/lsalcedo100/FusionFlux`
Archived at doi:10.5281/zenodo.22545928 (v0.4.0; the DOI for all versions is 10.5281/zenodo.22215142)

Abstract

Empirical confinement scaling laws are a principal input to the size a next-step tokamak has to reach. On the International Tokamak Physics Activity (ITPA) Global H-mode Confinement Database (HDB5, standard analysis set STD5; 6228 slices from 4471 discharges across 18 database device/wall labels), under cross-validation grouped by discharge, a random forest reduces log error by 29% against a power-law baseline refitted within the same folds and 36% against the published IPB98(y,2) reference. That margin does not survive the question a scaling law exists to answer. Under leave-one-label-out the ordering of the three primary fitted models reverses exactly: the forest is worse than that same power law on all 13 held-out database labels and on all 11 physical devices they reduce to. The reversal survives a nested hyperparameter search, a sweep of the row-count threshold, and repeated cross-validation over shuffled fold assignments. The forest's error tracks the Mahalanobis distance of the held-out label's mean log-feature vector from the training distribution ($\rho = +0.85$) where the power law's does not ($\rho = -0.06$), and its predictions cannot exceed the range of its training targets. At a size-ordered cut reproducing ITER's 1.82× jump in major radius, the tree ensembles score closer to a constant predictor than to the power law, and split-conformal intervals calibrated to 90% fall to 3% and 0% row-level coverage, at unchanged width. Two repairs improve extrapolation, both supplying long-range structure: a Connor–Taylor constraint surface, selected post hoc and so the lowest observed error among those tested rather than a prediction, and a Gaussian process whose linear component keeps trending. Across the model families tested, extrapolation failure tracks long-range saturation rather than flexibility alone. For ITER the IPB98(y,2) and random-forest point predictions differ by a factor of 8.3.

Keywords: tokamak energy confinement; confinement scaling; machine learning; extrapolation; cross-validation; dimensional similarity; uncertainty quantification; ITER

*Independent researcher, Montclair, New Jersey, USA. `liamsalcedo100@gmail.com`. ORCID: 0009-0001-5039-8147.

1 Introduction

A tokamak ignites when the alpha heating from its own fusion reactions covers everything the plasma loses. The energy confinement time τ_E sets that loss rate, which makes it a principal input to how large a next-step device has to be. Empirical scaling remains a principal tool for projecting it to a device that does not exist yet [1]: a power law in the engineering parameters,

$$\tau_E = C\, I_p^{a_1} B_t^{a_2} \bar{n}_e^{a_3} P^{a_4} R^{a_5} \epsilon^{a_6} \kappa^{a_7} M^{a_8}, \tag{1}$$

fitted by least squares across discharges from the multi-machine confinement database. In log space Eq. (1) is exactly a linear model, so fitting one is ordinary least squares on a log design matrix. The reference fit is IPB98(y,2) [1], a regression over the ITPA multi-machine database whose exponents were used to size ITER; that database is still maintained and republished [2].

What such a law is used for is an extrapolation, and a large one. ITER's major radius is 1.82 times the largest machine in the database, and SPARC [3] is designed for a toroidal field 2.1 times the largest recorded in it. The law is asked for a number at a point no row occupies, and that is the regime in which a fitted regression carries no guarantee at all.

Replacing Eq. (1) with a flexible regressor is an obvious move, and machine learning has been productive elsewhere in this field: disruption prediction [4], transfer of a disruption predictor to a device that supplied almost none of its training data [5], and real-time magnetic control [6]. Confinement scaling itself has been revisited with symbolic regression, which abandons the power-law form rather than fitting it [7], and with a neural residual learned on top of a frozen power law [8]. Measured conventionally the flexible models win. On the rows used here the random forest scores 0.128 in log-RMSE against 0.181 for a power law refitted inside the same folds on the same rows and features, a margin of 29%, and against 0.199 for the published IPB98(y,2) law, a margin of 36%.

The two comparators are not interchangeable and both are quoted throughout. IPB98(y,2) is a historical reference: it was derived in 1998 from version 2.8 of this database's predecessor (DB2.8) rather than from the DB5.2.3 revision analysed here [2]. That predecessor overlaps the present database substantially in devices and operating regimes, so for most folds below the machine being held out contributed to the data IPB98(y,2)'s exponents were fitted on. It is therefore not a fold-specific comparator that can be described as having been fitted without the held-out rows, and no claim in this paper rests on treating it as one. The power law refitted inside each fold is the like-for-like baseline, and the reversal below is stated against it. Every number in this paper is scored on the nine engineering features; the ten-column matrix of Sec. 3, which adds IPB98(y,2)'s own prediction to them, is reported there as a control and nowhere else.

Related work on this database. Four recent studies apply machine learning to the same ITPA data and are the closest prior work. Nam and Seo [9] take cross-machine transfer as their subject: they align features to correct the covariate shift between devices and use deep ensembles for uncertainty, validating on conditions the model has not seen, including JET-ILW and ITER. Gao *et al.* [10] apply variational and sampling-based Bayesian neural networks to STD5 together with individual JET, DIII-D and ASDEX Upgrade data, reporting improved accuracy alongside posterior uncertainty. Xiong *et al.* [11] tune random forests and two neural architectures by Bayesian optimisation on STD5 and report R^2 up to 0.990, with a feature-importance analysis separating spherical from conventional tokamaks. Wang *et al.* [8] freeze a power law and learn its residual, reporting improved stability under shifts defined by parameter ranges.

This paper asks a different question, and the difference is the protocol rather than the model. None of these works is the target of a criticism here; what any of them reports is what its split measures. That is the point. Where a score on this database is computed without holding out a whole device, it describes discharges from machines the model has already seen, and Sec. 5 shows that the ranking such a split produces can invert completely once the held-out unit becomes a whole device. We do not propose a better predictor. We ask whether the validation that selects one answers the question a scaling law exists for, measure what the honest answer costs, identify the property of a model that decides it, and test the repair that physics rather than flexibility supplies.

That measurement does not directly address the deployment problem of interest, for a reason about the data rather than about the models. Cross-validation on this database holds out *discharges*, and every machine in a held-out fold also appears in the training fold, so a held-out row has hundreds of near neighbours recorded on the same device at nearby settings. What the score quantifies is how much of JET is predictable from the rest of JET. That a validation split should match the generalisation being claimed, and that grouped observations flatter a model when it does not, is well understood in fields whose data arrives in clusters [12]; it is not what a commonly reported comparison on this database measures. The quantity a scaling law is built to supply is a prediction for a machine that does not yet exist.

This paper measures what the deployment-matched split costs. We escalate what is held out in four steps, a discharge, then a database label, then a physical device, then an entire size range, and score the same models under each. The results are: (i) under leave-one-label-out the ranking of the three primary fitted models is the exact reverse of their ranking under grouped cross-validation, and the best cross-validated model loses to a log-linear power law on all 13 held-out database labels and all 11 physical devices; (ii) three measurements of why, covering extrapolation distance, the training-target bound of a random forest, and what polynomial flexibility costs in the tail; (iii) a size-ordered split inside the database that reproduces ITER's extrapolation factor, at which conformal intervals calibrated to 90% cover 3% of rows without becoming any wider; (iv) the Connor–Taylor dimensional constraints, imposed as linear equality constraints on the exponents, giving the lowest observed error of any fit tested here at that cut, with no within-surface hyperparameter to tune, on a surface selected after those scores were seen; (v) a Gaussian-process experiment that separates flexibility from long-range saturation, holding the nonlinear component's specification fixed and varying only the mean-function specification, and corrects the diagnosis the earlier sections leave open; (vi) replication on rows the standard analysis set does not contain, in a second confinement regime, and on two scaling laws from biology, with a predictor-count ladder that separates the extrapolation failure, which persists at every predictor count tested, from the ranking inversion, which appears only once the flexible model wins the interpolation split; and (vii) a locked forecast for SPARC, JT-60SA and ITER. Section 3 begins with a property of the design matrix itself: it is rank deficient by two, and one of the two dependencies is that a published scaling law added as a model feature contributes nothing a log-linear fit can use.

2 Data and methods

Data. The ITPA Global H-mode Confinement Database, standard analysis set STD5 v5.2.3 [2], obtained from the Open Science Framework (OSF). The delivered file contains 6228 rows from 4471 discharges across 18 database device/wall labels, which correspond to 16 physical devices once the two JET entries and the two ASDEX Upgrade entries are each collapsed. The cleaning rule below requires finiteness and positivity in every used column; on this file it removes no rows, so all 6228 delivered rows are analysed. The target is the thermal confinement time `TAUTH`. No synthetic data appears in any result. The analysed file is pinned by SHA-256 and verified on load, so every number below refers to one specific set of bytes rather than to whatever the host currently serves.

Features and models. Nine log engineering features: plasma current, toroidal field, line-averaged density, loss power, major radius, elongation, inverse aspect ratio, effective mass and minor radius. The analytic IPB98 prior is *excluded* as a feature throughout the extrapolation arms. Its exponents were fitted on a predecessor of this database that overlaps it in devices and regimes, so for most folds it carries information about the held-out label; retaining it would import that overlap into every fold. The model set is a ridge regression on the log features (i.e. a fitted power law), a random forest [13] (300 trees), a histogram gradient booster, polynomial ridge controls of degree 2 and 3, and a mean baseline. IPB98(y,2) evaluated analytically is reported as a historical reference and is marked as such throughout: it is not refitted within folds, and its scores are not part of the ranking claim.

What is and is not tuned. Every model is fitted and scored in $\log \tau$: the target is $\log \tau$ and the reported error is the root-mean-square residual in the same variable. Exponentiation to τ enters only where a quantity is quoted in seconds, such as the mean absolute error and the device forecasts of Sec. 12. None of the models in Table 1 is hyperparameter-tuned, and "library defaults" is not a specification, so the consequential settings are given here in full. The power law is `StandardScaler` followed by `Ridge` with $\alpha = 1$ and the SVD solver, chosen before any held-out score was computed and never varied. The random forest is `RandomForestRegressor` with 300 trees, `criterion="squared_error"`, unlimited depth, `min_samples_split` = 2, `min_samples_leaf` = 1, `max_features` = 1.0 and bootstrap resampling. The gradient booster is `HistGradientBoostingRegressor` with squared-error loss, learning rate 0.1, at most 100 boosting iterations, 31 leaves, no depth limit, `min_samples_leaf` = 20, no L_2 penalty and 255 bins. Its `early_stopping="auto"` resolves to *off* at every sample size here, since that setting enables early stopping only above 10 000 rows and no training fold reaches that. The polynomial controls are `PolynomialFeatures` of degree 2 and 3 without a bias column, then the same scaler and ridge. Neither tree ensemble is standardised, which is immaterial to both.

Every estimator carries the seed 42, and the forest is fitted in parallel but predicts single-threaded so that reruns are bit-identical. Standardisation sits inside the pipeline, so it is refitted on the training rows of every fold and never sees a held-out row. No within-model hyperparameter search is performed on any reported cross-validation score, so nested tuning is not required for those numbers, and Sec. 5.2 reports what a nested search does when one is run anyway. Comparing several prespecified model families is still a selection of a kind, and two later choices go further: the constraint rung of Sec. 9 and the damping of Sec. 7 were both picked after seeing held-out scores, and both are flagged where they appear. The Gaussian processes of Sec. 13 do estimate kernel hyperparameters, by marginal likelihood on training rows alone.

Splits. (i) Grouped cross-validation by discharge: `GroupKFold` with five folds, grouped on the discharge identifier, so every slice from one discharge falls in the same fold. It is constructed with `shuffle=False`, its default, so the partition is deterministic and no seed enters it; Sec. 5.2 reports the same comparison over shuffled partitions. (ii) Holding out a whole unit, in two versions that are named separately throughout because the 18 labels are not 18 physical devices. *Leave-one-label-out* (LOLO in the tables) holds out each of the 13 database device/wall labels with at least 30 rows in turn, training on every other row in the database. *Leave-one-device-out* holds out each of the 11 physical devices those labels reduce to once JET and JET-ILW are merged, as are the two ASDEX Upgrade entries. The distinction matters here more than it usually would: holding out JET-ILW while JET stays in training does not hold out an unseen device, and this paper's own argument is that the held-out unit has to match the deployment being claimed. Leave-one-label-out is the default reported column because it is the finer partition and the one a practitioner would reach for first; every claim that turns on a device being genuinely unseen is stated against leave-one-device-out and labelled as such. The 30-row threshold governs only which labels are scored, not which are trained on, so the five labels below it (COMPASS, TCV, START, TDEV and TFTR, 43 rows between them) stay in every training fold; each fold therefore trains on 17 labels. It was fixed before any held-out score was computed, as the smallest round number giving a per-label log-RMSE over enough rows to be worth reporting, and it is not a tuned quantity: Sec. 5.2 sweeps it over 10, 20, 30 and 50 rows and reports what the win count does, and also reports what happens when the five sub-threshold labels are dropped from training as well. (iii) A size-ordered cut: labels are ordered by median major radius, the model is trained below a cut and scored on every label above it, and the reported cut is the one whose size ratio is closest in log to the ratio between this database and ITER. That rung has a ratio of 1.823, trains on 14 database labels and scores 2730 held-out rows, and is called the ITER-size-matched cut throughout. The match is to ITER's major-radius ratio and to nothing else: ITER is a burning plasma dominated by alpha self-heating at a gain no row here approaches, so clearing this bar is necessary and not sufficient.

The score is the root-mean-square error in $\log \tau$, $\mathrm{RMSE}_{\log \tau} = \sqrt{\overline{(\log \hat{\tau} - \log \tau)^2}}$, written log-RMSE here and in the figures. It is the plain log residual, not the $\log(1 + y)$ quantity usually called RMSLE; confinement times are strictly positive, so no offset is needed, and the name is avoided here to keep the two distinct. Intervals are percentile bootstraps resampling whole *units*, since the claim concerns behaviour on an unseen unit: 2000 resamples drawn with replacement over the 13 labels, seeded at 20240617, reported as the 2.5th and 97.5th percentiles. Paired intervals resample the per-label difference rather than the two error levels, which removes the shared unit difficulty. These intervals are bootstrap summaries over held-out units rather than confidence intervals under independent replication: any two folds share almost all of their training rows, so the units they score are not independent experiments. The descriptive win counts carry the claim; the intervals describe its spread.

Preprocessing. Rows are dropped only for non-finiteness or non-positivity in a used column, which on this file removes none of the 6228 delivered rows; no outlier rule, no winsorising and no reweighting is applied. Features are natural logs of the raw engineering columns, taken after cleaning. Minor radius is reconstructed as $a = \epsilon R$ at this stage, which is the origin of one of the two exact collinearities of Sec. 3. Wall variants (JET-C and JET-ILW, and the two ASDEX Upgrade entries) are separate labels in the database and are kept separate except where the text says devices are collapsed to 11.

Software. Python 3.12.14, scikit-learn 1.7.2, NumPy 2.2.6, SciPy 1.18.1 and pandas 2.3.3. The version of every library is pinned in the repository's `constraints.txt`. Any setting left at its library default therefore resolves to those versions' default; the consequential settings are written out above rather than referenced, because a default may differ in a later release.

3 The design matrix is rank deficient by two

Three column counts appear in this paper and they are not the same set. The models see nine log engineering features. Adding the analytic IPB98 prior to those nine gives the ten-column matrix examined here. The refit of Eq. (1) in Sec. 4 drops both redundant columns and uses nine, eight independent variables plus an intercept, which is why its condition number is a healthy 10.7.

Standardised, the ten-feature matrix has rank 8, with the last two singular values at 4.9×10^{-14} and 2.4×10^{-14}. The models keep the redundant nine-column representation, which is free for least squares but not obviously so for ridge, since a penalty is applied in the coordinates it is handed. Minor radius is a parameter a practitioner building this model would naturally supply, so nine columns is the set a reader would arrive at unprompted; promoting the eight-column variant because it turns out cleaner would be a selection on held-out scores of the kind Sec. 9 flags in its own result. The eight-column fit is therefore reported as a control rather than as the headline. Dropping `log_a_m` and refitting on eight independent columns leaves the log-linear scores unchanged to four decimals under all three splits, 0.181, 0.214 and 0.278, moves the tree ensembles by at most 0.035, and changes no ordering anywhere.

The estimator is not load-bearing either. Ordinary least squares on those same eight independent columns, carrying no penalty at all, scores 0.1812, 0.2141 and 0.2790 against the reported ridge's 0.1812, 0.2141 and 0.2778: identical to six decimals under cross-validation, apart by 0.0001 under leave-one-label-out, and by 0.0012 at the size cut. At $\alpha = 1$ on standardised columns with thousands of rows the penalty does almost nothing, so the ridge here is a numerically stable spelling of the least-squares power law rather than a regularised alternative to it. Neither the choice of estimator nor the redundant column carries any of the results below. Projecting candidate vectors onto the null space confirms two exact dependencies at relative residuals of order 10^{-16}: minor radius is derived as $a = \epsilon R$ during cleaning, and the IPB98 prior is a fixed log-linear combination of the other eight features.

The second is the substantive one. Adding a published physics scaling as a model feature feels like adding knowledge; in log space it provably is not, and it contributes exactly nothing to a log-linear fit. The deficiency raises no error, because `scipy.linalg.lstsq` inverts through the SVD pseudoinverse and returns the minimum-norm member of a two-parameter family of coefficient vectors that fit identically well.

Two aspects of the test for such a dependency are worth recording, since both give false negatives. The null space is a *subspace*, so when the deficiency exceeds one the basis vectors returned by the SVD generally do not resemble the dependency sought even when that dependency is exactly right; the correct test is $\|v - N^\top N v\|$. And standardisation rescales the null space, so a raw dependency must be mapped into standardised coordinates before comparison. Omitting either gives residuals of 0.41 and 0.96 for dependencies that are exactly true.

4 Refitted power-law exponents and weakly identified directions

Refitting Eq. (1) on all STD5 rows (design matrix 6228×9, condition number 10.7) gives I_p 1.08 against the published 0.93 and R 1.58 against 1.97, while P and B_t return almost exactly. This disagreement is expected: IPB98 was fitted to a selected ITER-relevant subset, whereas this fit applies no selection criteria and includes the spherical tokamaks. A refit on a different population is a different regression.

The exponents are not sensitive to how the least-squares problem is solved. Independent implementations by the normal equations, by QR and by the SVD pseudoinverse agree to 7.6×10^{-13}; a Cholesky solve of the normal equations is three orders of magnitude less accurate, since it forms $X^{\top}X$ and so works at the square of the condition number, which at $\kappa = 10.7$ is still far below anything that affects a reported digit.

Where the two differ matters more than by how much. Decomposing the difference between the exponent vectors along the singular directions of the design matrix, 77% of it lies in the single weakest direction, which carries 0.3% of the matrix's variance, while the three strongest directions carry 82% of the variance and account for 0.75% of the disagreement. That weak direction is plasma current traded against machine size, which is weak for a structural reason: tokamaks are not designed to vary the two independently, so the existing multi-machine design leaves that direction weakly identified. More shots at the operating points these machines already run would not separate it; shots deliberately placed to break the correlation could.

Why the size dependence in this database is weaker than IPB98(y,2)'s is itself an active question, and Hall *et al.* [14] attack it directly on DB5.2.3: they treat the multicollinearity among the predictors explicitly, and search for the smallest subset of points that most reduces the fitted major-radius exponent, reporting that the subset so identified is delineated in dimensionless space. A continuation of that work [15] sharpens the finding: the weak size dependence is attributed predominantly to a subset of discharges localised in dimensionless space, distinguished by normalised pressure, collisionality, normalised gyroradius and safety factor rather than by any single device or by metallic walls, and excluding it recovers a major-radius dependence broadly consistent with IPB98(y,2). That is a different question from the one asked here. They ask where the changing size dependence comes from; this paper takes the fitted exponents as given and asks how the choice of validation split changes which model is preferred and how far it can be trusted outside the data. The two bear on each other: if the size direction is partly an artefact of a localised subset, then the size-ordered cut of Sec. 6 is measuring transfer across a boundary those authors would draw differently, which is a caveat on how much weight that one axis can carry and not on the ranking reversal, which is measured per unit rather than per size.

5 The ranking inversion

Table 1 is the central result. Among the three primary models, a power law, a random forest and a gradient booster, the order under one split is the exact reverse of the order under the other. The random forest is the best of the three primary models under cross-

Table 1: The headline comparison under the conventional reporting choices: the same models and the same nine engineering features, scored one way per split (LOLO is leave-one-label-out over the 13 eligible labels). The two columns differ in more than the held-out unit, since they also differ in row population and in how per-row scores are aggregated, and the 13 labels are not 13 independent devices. Sec. 5.2 matches the population and the aggregation and repeats the comparison over physical devices; the inversion survives all of it. The rank columns cover only the six models above the rule, all of which are refitted inside every fold. IPB98(y,2) is set below the rule and carries no rank: it is not refitted within folds and is not part of the ranking claim.

Model	CV	LOLO	Ratio	CV rank	LOLO rank
random forest	0.128	0.465	3.6×	1	4
hist gradient boosting	0.130	0.359	2.8×	2	3
ridge, log-cubic	0.148	0.849	5.7×	3	5
ridge, log-quadratic	0.158	0.300	1.9×	4	2
ridge, log-linear	0.181	0.214	1.2×	5	1
mean baseline	0.869	0.994	1.1×	6	6
Historical reference, not fold-fitted					
IPB98(y,2), analytic	0.199	0.188	1.0×	n/a	n/a

validation and the worst of the three on a label it has not seen. It is not the best model in the paper: the Gaussian process of Sec. 13 scores 0.112. Both columns compare the same model specifications and the same features, but they differ in scoring population and in aggregation as well as in held-out unit; Sec. 5.2 matches those choices and shows the reversal survives. Against the fold-fitted power law the forest is 29% better in the CV column and 117% worse in the LOLO column. The analytic IPB98(y,2) row is the published reference: its exponents come from DB2.8, a predecessor of this database that overlaps it heavily in devices and regimes [2], so it is reported as a historical reference rather than as a model refitted within these folds. The two polynomial rungs are controls, scored in the same table but outside the ranking claim, and including them breaks the inversion rather than extending it: the log-cubic rung is third of the five fitted models under cross-validation and last of them on a held-out label.

Because the labels differ enormously in difficulty, the marginal bootstrap intervals overlap and are the wrong test; resampling the *paired* per-label difference removes that shared difficulty. The random forest is worse than the log-linear power law on **13 of 13** database labels, mean gap +0.251, 95% interval $[+0.157, +0.342]$. The gradient booster is worse on 12 of 13. Figure 1 shows the reversal together with the three extrapolation diagnostics of Sec. 5.1. Section 5.2 repeats this over the 11 physical devices, where the labels are no longer two eras of one machine counted twice, and the gap widens to +0.315 with an interval of $[+0.221, +0.408]$.

5.1 Three diagnostics of the failure

Distance. Ranking each model's per-label error against the Mahalanobis distance of that label's mean log-feature vector from the training mean gives $\rho = +0.85$ for the random forest and +0.54 for the gradient booster, against $\rho = -0.06$ for the log-linear power law. The random forest's error rises strongly with extrapolation distance and the gradient booster's trends upward more weakly. The power law's does not: its error is uncorrelated with how far the label sits from anything it saw.

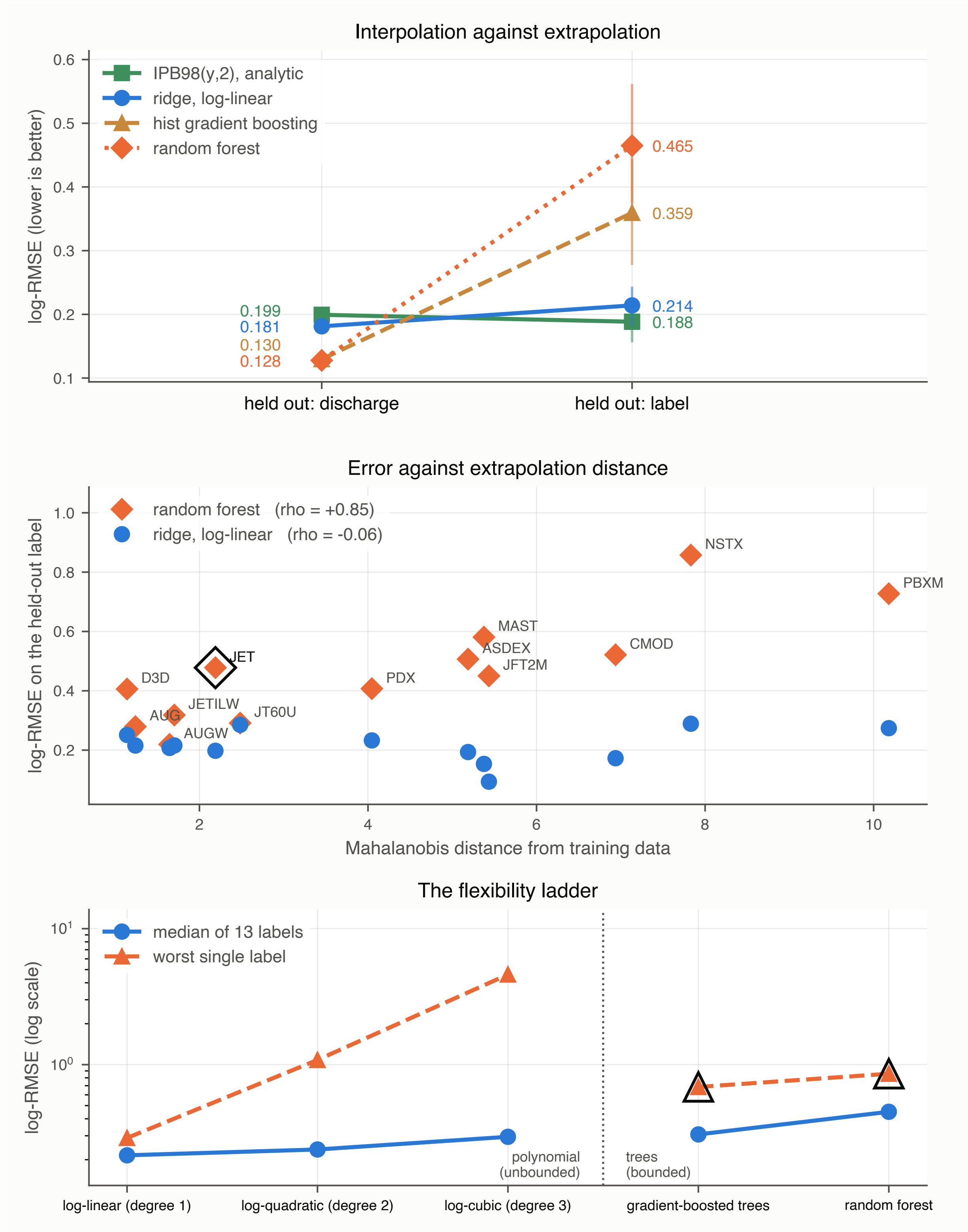

Figure 1: Interpolation against extrapolation. Top: each model under both splits, with bars giving the 95% interval over the 13 held-out labels; the lines crossing is the result. Middle: per-label error against Mahalanobis distance from the training data, rising strongly for the random forest, more weakly for the booster, and flat for the power law. The circled label is the second failure mode, where the true confinement times run above anything the random forest can output. Bottom: the flexibility ladder, where what grows with flexibility is the tail rather than the centre; the circled points are the two tree ensembles, both observed here to stay inside the training-target range, though only the random forest is provably bounded to it.

Table 2: Per-label log-RMSE under leave-one-label-out, ordered by Mahalanobis distance of the held-out label's mean log-feature vector from the training mean (computed through the pseudo-inverse, since that covariance is singular by Sec. 3). The trees degrade with distance; the power law does not.

Held out	rows	dist.	IPB98	ridge	hist GB	forest
D3D	388	1.1	0.252	0.251	0.246	0.406
AUG	1377	1.2	0.197	0.216	0.281	0.279
AUGW	767	1.6	0.218	0.207	0.212	0.219
JETILW	866	1.7	0.257	0.216	0.244	0.318
JET	1762	2.2	0.148	0.198	0.487	0.478
JT60U	100	2.5	0.275	0.285	0.307	0.291
PDX	97	4.0	0.227	0.233	0.323	0.407
ASDEX	431	5.2	0.131	0.194	0.207	0.507
MAST	39	5.4	0.136	0.154	0.317	0.581
JFT2M	69	5.4	0.095	0.094	0.234	0.450
CMOD	45	6.9	0.119	0.173	0.569	0.521
NSTX	185	7.8	0.222	0.289	0.686	0.857
PBXM	59	10.2	0.172	0.274	0.559	0.727

Thirteen points is few enough that each of those numbers needs an uncertainty. Permuting the distances 20000 times gives $p = 0.0007$ for the forest, $p = 0.060$ for the booster and $p = 0.85$ for the power law. That permutation treats the 13 database labels as exchangeable, which they are not exactly: two physical devices contribute two wall-era labels each, so these p-values are descriptive in the same way the sign test of Sec. 5.2 is. The same caveat applies to the coverage-against-distance correlations of Sec. 8. Recomputed over the 11 physical devices, which removes the duplicated wall-era labels, the forest's correlation weakens slightly to $+0.77$ ($p = 0.005$) and the power law's is unchanged at -0.06 ($p = 0.85$), so the contrast the section rests on survives the collapse. That permutation p-value remains descriptive for the same reason the sign test's does: the 11 leave-one-device-out fits share almost all of their training rows, so they are not 11 independent experiments. The booster's falls from $+0.54$ to $+0.38$ ($p = 0.25$). Neither figure clears a conventional threshold, and the booster's correlation is read throughout as suggestive rather than established. Dropping each label in turn moves the forest's ρ only within $[+0.81, +0.88]$, so no single extreme device carries it. One control changes how the rest should be read: the *mean baseline*, which fits nothing, also correlates with distance, at $+0.56$. Distance predicts how hard a unit is in general. What is unusual is not that the trees degrade with it but that the power law does not.

Mahalanobis distance between mean log-feature vectors is one measure of how far a unit sits from the training set, and it is the only one used here. It compares first moments through a single pooled covariance, so it is insensitive to differences in the shape or spread of a unit's operating envelope at a fixed mean, and the covariance it uses is singular by Sec. 3 and so is inverted through the pseudo-inverse. Every statement below about "distance" is a statement about that one diagnostic, not about distributional difference in general.

Table 2 gives the per-label scores behind that correlation. On the four most distant labels (PBX-M, NSTX, C-Mod, JFT-2M, all small, one of them spherical) the forest's error is 2.7 to 4.8× the power law's; on the four nearest it is within 1.7×.

A hard bound. A random forest predicts by averaging leaf values, and every leaf value is itself a mean of training targets, so every prediction it can make lies in $[\min y_{\text{train}}, \max y_{\text{train}}]$ whatever the features say. With JET held out, 48% of its rows exceed the maximum confinement time anywhere in the training fold, and its largest held-out target is 3.7× above anything any tree in the forest can output. More rows would close this only if they carried larger targets: within the same target range no amount of additional data, and no tuning, moves a ceiling that is a property of the functional form.

A gradient booster carries no such guarantee. It sums tree outputs onto an initial estimate rather than averaging targets, and nothing in that construction confines the sum to the training range, so the argument above does not transfer to it. What we can report is that it never uses the freedom: over the 13 leave-one-out folds and the size cut of Sec. 6, its largest prediction stays below $\max y_{\text{train}}$ on every one, coming closest with JET held out at 0.066 in logs beneath the ceiling and sitting 0.567 beneath it across the size cut. Its boundedness on these rows is measured rather than guaranteed, and every claim below that needs the guarantee is made about the forest.

Flexibility costs the tail, not the centre. A polynomial ladder in the log features (degrees 1–3) is flexible but still extrapolates. Degree 2 is *not* meaningfully worse than degree 1 on a typical label: median 0.238 against 0.216, better on 8 of 13, paired interval $[-0.013, +0.237]$ containing zero. What flexibility costs is the tail. Degree 1's worst label is 0.289; degree 2's is 1.083 and degree 3's is 4.601, both on C-Mod. A bounded range is simultaneously why neither ensemble extrapolates and why neither blows up: their worst cases, 0.686 and 0.857, sit far below degree 3's.

Read together, these three diagnostics invite a summary this paper does not endorse. Every flexible model scored so far is also either bounded, like the two ensembles, or divergent, like the cubic, so nothing above can tell apart "flexible models fail out of distribution" from "models that stop trending outside their data fail out of distribution". The two make identical predictions about everything in Table 1. Section 13 builds the model that separates them and supports the second interpretation, so the reader should carry the flexibility reading as provisional from here rather than as the paper's conclusion.

5.2 Sensitivity to population, aggregation, and device definition

Table 1 attributes a difference to the split, so anything else that differs between its two columns is a confound. Three things do. Grouped CV scores all 6228 rows while leave-one-out scores only the 13 labels large enough to hold out. Grouped CV pools out-of-fold predictions into one log-RMSE over every row, where JET and AUG dominate, while leave-one-out averages per-label scores, where every label counts once. And JET and JET-ILW are one physical tokamak either side of a wall change, as are AUG and AUGW, so holding out JET-ILW while training on JET does not hold out an unseen device.

The labels too small to score. A fourth choice is worth stating because it looks like a confound and is not. Five labels hold fewer than 30 rows (COMPASS, TCV, START, TDEV and TFTR, 43 rows between them, under 1% of the database) and none can carry a held-out score of its own. They are nonetheless in every training fold, since the threshold governs scoring rather than training. They are also among the most unusual points in the feature space, so whether a flexible model can exploit those few rows is a fair question to ask of the ranking.

Table 3: The random forest and the power law under both row populations, both aggregations, and both definitions of a machine. The inversion is the pattern of the random forest leading under cross-validation and trailing the power law under leave-one-out; it holds in all eight combinations.

	pooled over rows		each unit equally	
Split	forest	ridge	forest	ridge
CV, all 18 labels	0.128	0.181	0.149	0.238
CV, the 13 scored labels	0.123	0.177	0.124	0.187
leave-one-label-out (13)	0.410	0.214	0.465	0.214
leave-one-device-out (11)	0.551	0.200	0.527	0.212

Dropping them from training as well answers it. Every fitted model gets slightly worse without them, by 0.012 for the power law, 0.027 for the gradient booster and 0.007 for the random forest, and the ordering over the 13 held-out labels is unchanged: power law 0.214 to 0.226, gradient booster 0.359 to 0.387, random forest 0.465 to 0.472. The 43 rows help the tree ensembles slightly, as one would expect of rows in a sparse region, and nowhere near enough to recover the inversion. The analytic reference is unaffected, since it fits nothing.

Table 3 varies all three. The forest leads the power law under every cross-validation cell and trails it under every leave-one-out cell, so the inversion is a property of the split rather than of the population or the aggregation. Collapsing the variants makes the forest's failure worse rather than better: over 11 physical devices its mean gap over the power law grows from +0.251 to +0.315, with a paired 95% interval of $[+0.221, +0.408]$ against $[+0.157, +0.342]$ over the labels, and it loses on 11 of 11. The published law's score is almost unmoved by any of it, 0.188 to 0.184 per unit.

A newer published law. IPB98(y,2) is the ITER reference but not the most recent scaling fitted to this database family. ITPA20 and ITPA20-IL [2] were derived from DB5.2.3 and weaken the size dependence considerably, so they are the obvious second reference. Evaluated analytically on the rows analysed here, ITPA20 scores 0.265 over all rows, 0.267 per label and **0.272** across the ITER-size-matched cut; ITPA20-IL scores 0.393, 0.446 and 0.363. Both are worse than IPB98(y,2) on this cleaning, and two caveats belong with that: each was fitted to a different selection out of DB5.2.3, and each specifies an elongation this database does not supply under that definition, so their absolute level here is not a fair test of either law. What survives both caveats is the comparison this paper is about. At the ITER-size-matched cut ITPA20 lands at 0.272 against the random forest's 0.938 and the gradient booster's 1.072, factors of 3.4 and 3.9, and essentially level with the fold-fitted power law's 0.278. Despite the definition mismatches, both published power-law forms score substantially below the saturating tree models at that cut. Like IPB98(y,2), neither is refitted within these folds, and ITPA20 and ITPA20-IL were fitted to selections out of the very revision analysed here.

Where the row threshold is set. The 13-of-13 count is reported at a 30-row eligibility threshold, which is a choice, so it is swept. Table 4 scores leave-one-label-out at 10, 20, 30 and 50 rows. Only which labels are *scored* changes: every label stays in every training fold at every threshold, so the models being compared do not change. The power law wins every scored label at 20, 30 and 50 rows, and 14 of 15 at 10 rows, where the exception is TCV. TCV carries 11 rows, which is why it sits below the threshold in the first place, and a per-label log-RMSE over 11 rows is not a quantity this paper would report on its own. The mean gap moves between +0.211 and +0.251 across the four thresholds and the sign test stays below 10^{-3} throughout, so the headline does not depend on where the line is drawn.

Table 4: Leave-one-label-out at four eligibility thresholds. Only the set of labels scored changes across rows; the training folds and the models are identical throughout. The 30-row threshold is the one reported elsewhere in this paper. p is the exact two-sided sign test on the win count.

min. rows	labels scored	power law	forest	forest worse on	p
10	15	0.262	0.473	14 of 15	9.8×10^{-4}
20	13	0.214	0.465	13 of 13	2.4×10^{-4}
30	13	0.214	0.465	13 of 13	2.4×10^{-4}
50	11	0.223	0.449	11 of 11	9.8×10^{-4}

One partition, or any partition. The cross-validation column comes from a single deterministic `GroupKFold` assignment of discharges to five folds. Since this paper is about validation protocol, whether the 29% margin is peculiar to that one partition is a fair question. Repeating the comparison over ten shuffled assignments of the same discharges into the same number of folds gives a mean forest advantage of 27.8% over the power law, ranging from 27.0% to 28.5%, with the forest ahead in all ten. The deterministic partition's 29.5% sits just above that range, so the reported margin is if anything mildly favourable to the forest, and the interpolation win it later gives up is a property of the comparison rather than of one fold assignment.

Tuning the flexible models. Every score above uses the prespecified, untuned settings of Sec. 2, which leaves open whether the ensembles lost only because they were not hyperparameter-tuned. We tested that by giving both a grid search nested inside each training fold: three grouped inner folds by discharge over the training rows only, so the held-out discharge fold, label or size cut never takes part in choosing a hyperparameter. The forest searched `max_features` over $\{1.0, 0.5, \mathrm{sqrt}\}$ and `min_samples_leaf` over $\{1, 5\}$, the booster `learning_rate` over $\{0.05, 0.1\}$ and `max_leaf_nodes` over $\{15, 31\}$.

The search does real work on the forest and changes nothing about the conclusion. It moved away from the baseline `max_features` = 1.0, `min_samples_leaf` = 1 setting in 13 of the 19 folds, and it helps where the paper says a flexible model should help: cross-validation improves from 0.128 to 0.126 and the leave-one-label-out mean from 0.465 to 0.403. At the ITER-size-matched cut it makes the forest *worse*, 0.938 to 1.121, against the power law's 0.278. The booster's search returned scikit-learn's defaults in all 19 folds, so its numbers are unchanged.

The inner folds above group by discharge, which is how a practitioner would ordinarily tune. Grouping them by whole machine instead matches the way the model is deployed, and it helps both models: the forest falls to 0.376 and the booster to 0.350. Both remain well above the power law's 0.214, so the reversal survives either selection procedure.

Tuning therefore widens the interpolation margin that later inverts and does not repair the extrapolation failure, which is what the training-target bound predicts: no setting of these hyperparameters moves a ceiling that comes from averaging training targets.

The win counts support an exact test, under a null that is worth naming precisely. Call it the independent-unit-sign null: each unit, label or device as the case may be, is assigned to either model independently with equal probability. Under it, losing 13 of 13 has two-sided $p = 2.4 \times 10^{-4}$ and losing 11 of 11 has $p = 9.8 \times 10^{-4}$. The device-collapsed count is the one to quote, since the 13 database labels are not 13 independent devices, and the same non-independence is a reason to prefer these counts to the 13-label bootstrap.

That null is an idealisation, and in one respect a generous one. The leave-one-device-out fits are not independent experiments: any two of them share all but two of their training machines, so the per-device outcomes are correlated through the training set even though the held-out devices are distinct. The p-values above should therefore be read as a compact summary of how lopsided the win count is, not as the output of n independent replicates. The descriptive statement carries the claim on its own: the forest is worse on every physical device scored, and the paired interval on the per-device difference excludes zero.

6 A size-extrapolation stress test matched to ITER

Holding out JET still leaves the rest of the database spanning much of its range, so leave-one-out extrapolates in identity while interpolating in size. ITER's major radius is 6.2 m against 3.40 m for the largest row here, a factor of 1.82. That factor is available inside the database: training on the 14 smallest database labels (up to DIII-D, 1.865 m, 3498 rows) and predicting the 4 largest labels (up to JT-60U, 3.40 m, 2730 rows) demands a ratio of 1.823 against ITER's 1.824. The cut is selected by proximity to that ratio rather than by eye, so it is a property of the data.

Table 5 places that cut beside the two easier splits, and Figure 2 shows the whole size-ordered sweep. At the cut the power law scores 0.278 against the analytic law's 0.194 and a constant predictor's 1.459. The random forest scores 0.938 and the gradient booster 1.072, so both sit nearer the constant than the power law in absolute error. Asked the question a scaling law exists to answer, the two tree ensembles that lead the conventional cross-validation comparison land closer to a constant than to the published law they beat by 36% under cross-validation. The ordering is identical on all three individually scored held-out labels, so the pooled number is not an artifact of JET's weight. Four labels sit above the cut but only three are scored individually: TFTR contributes 2 rows, far below the 30 required to report a label on its own, and enters only the pooled number. Dropping the spherical tokamaks moves the random forest from 0.938 to 0.936, so the effect is not driven by including the spherical devices; other shape or regime covariates that track size are not ruled out by this control.

Table 5: Three questions of increasing difficulty, scored in log-RMSE. The *size cut* column is the ITER-size-matched cut. Reading down it rather than across the table is the point: the two models that win the first column move far toward the constant baseline in the third, landing closer to it than to the power law in absolute error.

Model	discharge CV	LOLO	size cut
IPB98(y,2)	0.199	0.188	**0.194**
ridge, log-linear	0.181	0.214	**0.278**
random forest	0.128	0.465	**0.938**
hist grad. boosting	0.130	0.359	**1.072**
mean baseline	0.869	0.994	1.459

7 A power law with a bounded correction

The diagnosis in Sec. 5.1 suggests a specific repair. Fit the power law, learn a correction on its log residuals, and damp it by λ:

$$\log \tau = \underbrace{f_{\text{ridge}}(x)}_{\text{unbounded, log-linear}} + \lambda \underbrace{g(x)}_{\text{correction on residuals}} . \tag{2}$$

At $\lambda = 0$ this is exactly the ridge row of Table 1. Anchoring a learned correction on a frozen power law is also the design used in concurrent work on this database [8], which reports improved stability under parameter-defined distribution shift. This section makes no claim on that idea. What is varied here is the one property the diagnosis of Sec. 5.1 points at, the long-range behaviour of the correction, and the surrounding contribution is the device-level validation reversal, its mechanism, the size-matched stress test, the constrained-exponent comparison and the interval collapse, none of which is a residual-learning result. The two correction families are chosen to contrast the long-range behaviour the diagnosis implicates. They differ in other ways as well, and the controlled version of this comparison is Sec. 13.1. One is a degree-2 log polynomial, which is unbounded; the other is depth-2 boosted trees, which as measured below never leave the residual range they were trained on. The second is an *empirically* bounded corrector rather than a provably bounded one, for the reason given in Sec. 5.1. Clipping g to the range of training residuals closes that gap at no cost: the clipped variant replaces $g(x)$ by $\min(\max(g(x), \min_i r_i), \max_i r_i)$, where r_i are the base model's training residuals on that fold. Built standalone with and without the clip, the two score identically under all three splits, 0.150, 0.249 and 0.189, and the clip binds on none of the held-out rows across the size cut. Those absolute levels are not the swept hybrid's, since the standalone build fixes $\lambda = 1$ and rebuilds the pipeline rather than reusing the sweep; the claim is the paired comparison, and it is that the empirical bound and the provable one pick out the same model here. The mechanism below can be read either way.

Specification of the sweep. The damping factor is swept over nine values, $\lambda \in \{0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.8, 1.0\}$, chosen so that both endpoints are interpretable models rather than arbitrary rungs: at $\lambda = 0$ the hybrid is exactly the ridge row of Table 1, and at $\lambda = 1$ the correction is undamped. The base term is `StandardScaler` followed by `Ridge` with $\alpha = 1$ and the SVD solver, identical to that row. The polynomial corrector is `PolynomialFeatures` of degree 2 without a bias column, then the same scaler and `Ridge` with $\alpha = 1000$; the boosted corrector is `HistGradientBoostingRegressor` with `max_depth` $= 2$, learning rate 0.05, 200 boosting iterations, L_2 penalty 10.0, early stopping off, and seed 42. Both correction families carry deliberately strong penalties so that λ rather than the corrector's own regularisation does the damping, and neither corrector's hyperparameters are tuned. The whole estimator, base and correction together, is refitted from scratch inside every training fold of every split, and both stages are fitted on that fold's training rows. No rung is selected on a held-out score for the reported comparison: the sweep is reported in full, and where a single rung has to be picked, Sec. 7 states which criterion picked it and what the alternative would have given.

Across the ITER-size-matched cut the two go opposite ways. Both start at 0.278; the polynomial correction climbs to 0.356 and the bounded correction falls monotonically to **0.206**, better than plain ridge on all three held-out labels and by the widest margin on JT-60U, the most distant (0.440 to 0.281). Among the models scored at that cut it has the lowest held-out error of anything fitted without the held-out rows, 26% below plain ridge and 4.6× below the random forest, and within 6% of IPB98, which is a historical reference rather than a fold-fitted competitor. The constrained fit of Sec. 9, introduced later, is lower still.

The mechanism is measurable. Fitted on the 14 small labels, the base power law is *biased* on the held-out ones: mean log residual −0.218, so it over-predicts by about 20%, while its scatter (0.172) is no worse than in-sample (0.187). The tree correction never leaves the range it was trained on, on any held-out row, and inside that bound it points the right way and is largest where the bias is largest. The same bounded range that makes these ensembles useless as predictors of τ on a larger machine limits how much extrapolation error the correction can contribute, because the quantity they are bounded on is no longer a target that grows with machine size but a residual centred on zero. Bounded is not the same as right: a correction confined to that range can still point the wrong way, and what is measured here is that on these rows it does not.

Two limits should be stated, and they qualify every later summary of this section. Cross-validation selects $\lambda = 1$ in both families, while the rung best on a held-out label is $\lambda = 0$; nothing in the CV signal points at it, so a practitioner tuning on the split they would ordinarily use does not arrive at the rung that transfers. And scored at every rung of the size sweep rather than only the matched one, the bounded hybrid wins at 5 of 8 well-powered cuts, not all, with no rule over those eight points separating wins from losses. The claim that survives is narrow: at the cut matching the jump to ITER, the bounded hybrid equals the power law at $\lambda = 0$, where it is that model, and beats it at all eight non-zero damping settings swept there, with the mechanism measurable.

8 Calibrated intervals and their collapse

For a next-step device the point error is not the deliverable; the interval is. We calibrate split-conformal intervals [16] on log residuals, holding back 25% of *discharges* to calibrate.

The construction is specified as follows, so that a reader can reproduce the coverage numbers rather than infer the procedure from them. Within each training fold, the distinct discharge identifiers are permuted with `numpy.random.default_rng` seeded at 20240811, and the first round$(0.25\, n_{\text{discharges}})$ of the permutation are the calibration discharges; every row of a calibration discharge is a calibration row, and the model is fitted on the remaining rows only. Arms with fewer than 100 calibration rows are not reported. The conformity score is the absolute log residual $|\log \hat{\tau} - \log \tau|$. The half-width is the $\lceil (n+1)(1-\alpha) \rceil$-th smallest score in ascending order, at $\alpha = 0.10$, which is the finite-sample conformal rank rather than the plain empirical quantile; the +1 accounts for the test point itself being one

Table 6: Empirical coverage of nominal 90% split-conformal intervals, and the multiplicative half-width under leave-one-out.

Model	CV	LOLO	ITER cut	width
IPB98(y,2)	90%	89%	88%	1.40×
ridge, log-linear	90%	83%	70%	1.33×
bounded hybrid	90%	64%	76%	1.26×
hist grad. boosting	91%	45%	**0%**	1.23×
random forest	91%	35%	**3%**	1.23×

of the $n+1$ exchangeable scores, and where that rank exceeds n the half-width is reported as infinite rather than as a finite but invalid number. Ties need no rule, since the quantile is a rank on a sorted array. The interval is symmetric in $\log \tau$ and is a pointwise interval for a new observation, not for a latent mean. Under grouped cross-validation the per-fold coverages are pooled over rows; under leave-one-label-out each held-out label is calibrated and scored on its own fold and the coverages are then averaged over machines.

Holding out whole discharges keeps a calibration row and a test row from coming out of the same shot, but the conformity unit is still the row, and rows arrive in correlated clusters of quasi-stationary slices from one discharge. The standard finite-sample guarantee assumes exchangeable conformity scores and clustered rows do not supply that, so every coverage number below is empirical row-level coverage rather than a guaranteed rate. A discharge-level conformity score, or a cluster-aware conformal method, would be the way to recover a guarantee, and neither is used here.

That proviso is what the section measures. Empirical coverage is near nominal under grouped CV, where every model lands within a point of 90%, and that control is what licenses reading the rest of Table 6. It is a useful in-distribution control rather than a verified guarantee: clustered row-level conformity scores do not satisfy the finite-sample exchangeability condition either, so what the CV column shows is that the construction behaves as intended when the test rows come from labels the model has seen. The exchangeability assumption is no longer justified when the test rows come from a systematically held-out label absent from the model's training and calibration distribution, and what that costs empirically is measured below. That exchangeability is the binding assumption, and that repairing it requires knowing the shift, is the subject of the weighted-conformal literature [17], and calibrated uncertainty degrading under dataset shift is a general finding rather than a fusion one [18]. The random forest's 90% interval then covers 35% of rows on an unseen label and 3% across the ITER-size-matched cut; the gradient booster covers none of the 2730 rows at all.

The widths barely move: no model's interval changes width by more than 1.5% between the two arms, because the half-width is set by calibration rows drawn the same way in both. The intervals do not become vague out of distribution. They stay the same size and miss.

Coverage collapses along the same axis the errors grow, but only for the trees: $\rho = -0.77$ for the random forest and -0.54 for the gradient booster against $+0.27$ for the power law, mirroring the per-label error correlations of Sec. 5.1 with the sign flipped. The power law's intervals are somewhat too narrow everywhere rather than catastrophically too narrow far away, and for a next-step device that distinction is most of what matters, because distance is the one thing known in advance. None of this is a defect in conformal prediction. It is a measurement of an assumption being false.

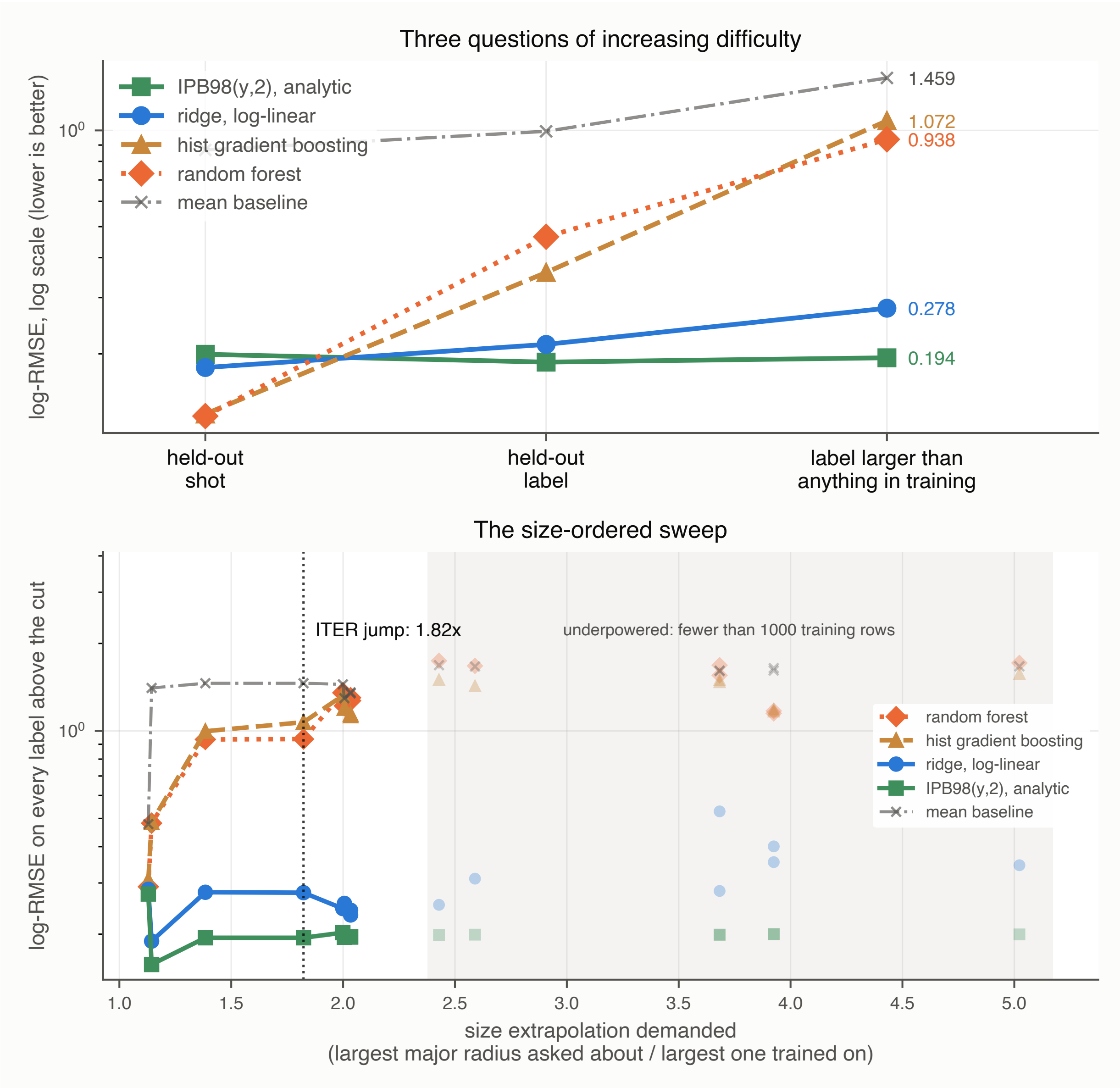

Figure 2: Size-ordered extrapolation. Top: the escalation across three splits of increasing difficulty, where the tree ensembles collapse toward the mean baseline. Bottom: the sweep across every cut, with underpowered cuts drawn faint and excluded from claims. The power law holds across the sweep and the trees do not recover.

9 Dimensional constraints from Connor–Taylor similarity

Every model to this point learns its functional form from the data. None is told any physics. That is worth revisiting, because the field has known since Connor and Taylor [19] that a confinement scaling law is not free: requiring it to be expressible in the dimensionless plasma variables imposes *linear equality constraints on the exponents*. A physics assumption is therefore a matrix C, and the fit becomes

$$\min_b \|Xb - y\|^2 \quad \text{subject to} \quad Cb = d, \tag{3}$$

which the Karush–Kuhn–Tucker (KKT) system solves in closed form. Nothing new is needed to run the experiment: the solver is the one written by hand for Sec. 4.

We derive the constraints in code from the definitions of ρ^*, β and ν^* rather than transcribing exponents from a paper, which makes the derivation checkable against something external. Written as exponents on (length, field, temperature, density),

$$\rho^* \sim T^{1/2} B^{-1} L^{-1}, \qquad \beta \sim n\,T\,B^{-2}, \qquad \nu^* \sim n\,L\,T^{-2}, \tag{4}$$

and a law expressible in whichever of these a model holds fixed must be invariant under every scale transformation that leaves those groups unchanged. Each such transformation contributes one linear equation in the exponents, so the admissible set is a surface and the number of constraints is set by how many groups are fixed:

constraint	groups held fixed	rows of C
Kadomtsev [20]	ρ^*, β, ν^*	1
collisionless	ρ^*, β	2
electrostatic	ρ^*	3

Fixing fewer groups admits more transformations and therefore imposes more constraints, which is why the hierarchy runs the way it does. C and d are built by `dimensional.constraint_matrix` from Eq. (4) alone, and their numerical rows are in `results/dimensional.json`.

Constraining a confinement scaling this way is not new, and IPB98(y,2) is itself an instance of it: the high-β Kadomtsev constraint was enforced when it was fitted, alongside a magnetic-field exponent fixed at 0.15 [2]. Its exponents accordingly land on the Kadomtsev surface [20] derived here at a distance of 0.00096, and on the collisionless surface at 0.0045. The first of those is expected rather than surprising, and we use it only as a check that the surfaces derived in code from Eq. (4) agree with the constraint the published law was fitted under. What is new here is not the idea of constraining the exponents but the comparison: a hierarchy of such surfaces, fitted and scored under a validation protocol matched to the deployment problem. Of the three rungs tested, the collisionless one gives the best size-cut transfer, and that rung was selected after those scores were seen.

At the ITER-size-matched cut the collisionless power law scores **0.183**, the lowest held-out error of any model fitted here without using the held-out rows (Table 7). The fit is blind to the held-out rows; the *search* is not. No coefficient ever saw a held-out row, but the constraint rung was chosen after seeing how each of the three scored at this cut, so the cut took part in model selection even though it never entered a fit. This is therefore the lowest error among the surfaces examined, not a prospective demonstration that the

Table 7: Fitting under the Connor–Taylor constraint hierarchy. The only difference between the first and last rows is the constraint: same features, same solver, no hyperparameter.

Model	in sample	CV	LOLO	ITER cut
power law, collisionless	0.1818	0.182	0.206	**0.183**
IPB98(y,2), analytic*	n/a	0.199	0.188	0.194
bounded hybrid (Sec. 7)	n/a	0.151	0.246	0.206
power law, electrostatic	0.2245	0.225	0.241	0.237
power law, Kadomtsev	0.1808	0.181	0.210	0.254
power law, unconstrained	0.1808	0.181	0.214	0.279

*historical reference, fitted to the DB2.8 predecessor of this database [2] and not refitted within these folds.

collisionless surface is the right one. It beats the bounded hybrid of Sec. 7, which was built specifically for that cut, and it beats the analytic law, which was fitted under a Kadomtsev constraint of its own on a predecessor database. It has no hyperparameter and no within-surface hyperparameter to tune, though the surface itself was selected post hoc from the three tested. Figure 3 shows the trade between what each constraint costs in sample and what it buys at the size cut.

The two constraints behave differently. The Kadomtsev constraint is *free*: at 0.1808 in sample it is identical to the unconstrained fit to four decimals, because the data already satisfies it unaided. It nonetheless beats the unconstrained fit at all 8 well-powered cuts, and descriptively at all 15 cuts in the size sweep when the underpowered ones are included, since a constraint the data satisfies on average still stops the fit wandering when the training half is small. The collisionless constraint costs 0.001 in sample and wins at 8 of 8 well-powered cuts, where a cut is called well-powered when it trains on at least 1000 rows. Push one rung further to a beta-independent law and it degrades sharply, to 0.237. There is an optimum in the middle of the hierarchy, and nothing in this analysis predicted where it would fall; as with the hybrid, cross-validation does not select it. Every constrained fit scored slightly worse than the unconstrained one under the cross-validation used here, so CV supplies no signal pointing at the constraint that transfers best. That is an observation about these folds rather than a theorem: a hard equality constraint can only match or worsen the *training* residual, and nothing forces it to worsen a held-out score.

It matters that this is a constraint and not a prior. Supplying the same physics as a Gaussian prior instead, shrunk along the weak singular direction of Sec. 4, is worth much less. That comparison sweeps three shrinkage geometries (isotropic, spectral and truncated) over the same nine penalty strengths $10^{-3}, 10^{-2}, \ldots, 10^{5}$, tabulated in full in `results/dimensional_prior_sweep.csv`. Targeting the shrinkage beats isotropic shrinkage at every one of those matched penalty strengths, which is a real effect, but nothing in that family beats simply adopting IPB98's published exponents outright. The hard constraint used here restricts the fit to a surface the Connor–Taylor similarity assumptions permit, where the Gaussian prior tested here favours a location and a direction in parameter space, and in this experiment the surface transfers better. That is a statement about these two families rather than about priors in general: a prior can favour a subspace, and approaches a hard constraint as its variance goes to zero.

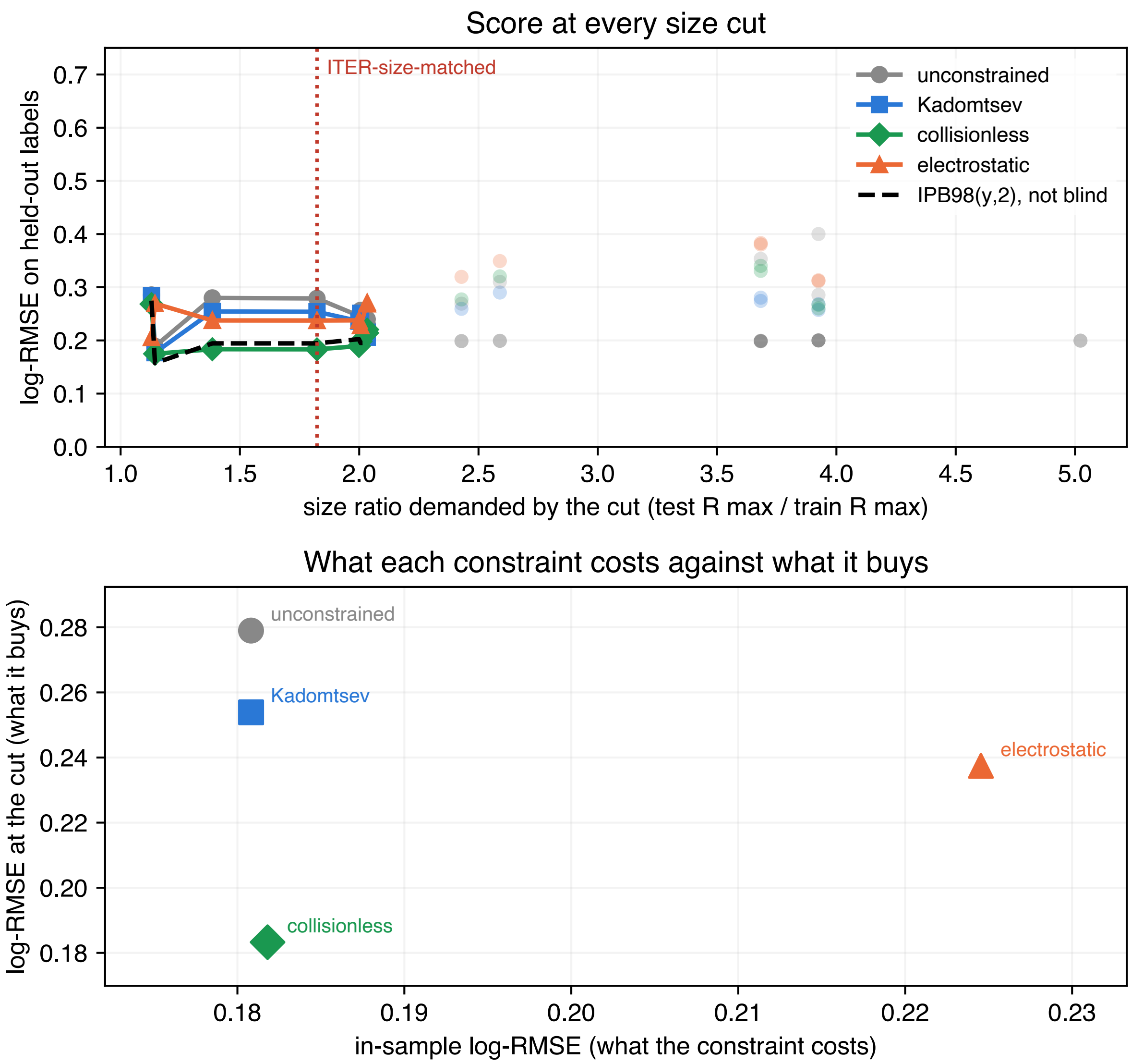

Figure 3: Cost against benefit across the size sweep. Top: the score at every size cut for the constrained and unconstrained fits, with underpowered cuts drawn faint and excluded from claims. Bottom: what each constraint costs in sample against what it buys at the ITER-size-matched cut. The constraint costs almost nothing where the data is plentiful and pays where it is not, and more physics is not monotonically better: the hierarchy has an optimum in its middle, at the collisionless rung. That rung was selected after these size-cut scores had been seen, so it is the best of the three surfaces examined rather than a prospective prediction.

10 Repairing the intervals, and the limit of the repair

The diagnosis in Sec. 8 makes a testable prediction. If the interval collapse is really about exchangeability rather than about the models, then calibrating on held-out *labels* instead of held-out discharges should repair it, and should repair it only as far as that unit reaches. Both halves are tested below.

The label-level scheme uses an inner leave-one-label-out loop, and nothing in it touches the target label. It is called label-level rather than machine-level deliberately: the calibration unit is the database device/wall label, so for a target of JET-ILW the calibration set still contains JET. The device-level variant that removes that overlap is reported at the end of this section. For each target m: remove m; the remaining labels are the training set. Inside that training set, hold out each label k with at least 30 rows in turn, fit on the others, and record $|\log \hat{\tau} - \log \tau|$ on k together with each of its rows' Mahalanobis distance from that inner fit set. Pool those residuals into the calibration quantile. Then fit the final model on all the training labels and score m once. Every conformity score therefore comes from a label its own model had not seen, which is the unit the test rows differ by, and no row from m enters any fit or any quantile. The distance-scaled variant fits $\sigma(d) = \exp(c_0 + c_1 d)$ to those calibration residuals by least squares on $\log|r|$ and divides the scores by it.

Neither scheme is a finite-sample-valid group-conformal procedure, and they are not claimed as one. Two things stand in the way. The conformity score is still a row, not a label, and rows arrive in correlated clusters, so the exchangeability the split-conformal proof needs is no more available here than in Sec. 8. And in the distance-scaled arm the scale function $\sigma(d)$ is estimated from the same calibration residuals it then normalises, which the standard argument does not cover. What these are is empirical held-out-label residual calibration: the quantile is the finite-sample conformal rank applied to those scores, and the coverage reported for it is measured rather than guaranteed. Recovering a guarantee would need label-level conformity scores and a scale model fitted on separate rows or cross-fitted, and neither is done here.

Both arms behave as the diagnosis predicts. On a held-out label, empirical row-level coverage returns close to nominal: every model lands within two points of it, tree ensembles included, and the random forest goes from 35% to 88%. Across the size cut none does. Neither recalibration procedure tested restores near-nominal coverage there, because every calibration unit is smaller than every test unit, and scaling the interval by extrapolation distance recovers only part of the gap (3% to 40% for the forest). A repair that stops exactly where the diagnosis says it should is stronger evidence for the diagnosis than one that worked everywhere would be.

The same calibration over physical devices. The scheme above calibrates on database labels, so for a target of JET-ILW the calibration set still contains JET. Collapsing the wall variants and repeating the whole procedure over the 11 physical devices removes that overlap: training, calibration and scoring then all use the device as the unit.

Table 9 says the diagnosis holds and the repair is less complete than the label-level numbers suggest. Plain split conformal is worse on a genuinely unseen device than on an unseen label, 31% against 35% for the forest and 38% against 45% for the booster, which is what removing the within-device overlap should do. Calibrating by device then recovers most of the shortfall for the forest (84%, and 88% with distance scaling, against 88% either way at label level) but noticeably less for the booster, which reaches 77% and 80% where

Table 8: Empirical coverage of nominal 90% intervals under three calibration schemes, on a held-out database label (LOLO) and across the ITER-size-matched cut. The device-level counterpart is Table 9.

	split conformal		by label		+ distance	
Model	LOLO	cut	LOLO	cut	LOLO	cut
random forest	35%	3%	**88%**	29%	88%	40%
hist grad. boosting	45%	0%	**90%**	4%	89%	13%
ridge, log-linear	83%	70%	91%	77%	92%	82%
power law, collisionless	85%	91%	91%	**95%**	92%	92%
IPB98(y,2)*	89%	88%	89%	89%	89%	87%

Table 9: The same three schemes with wall variants collapsed, so every calibration unit is a physical device and the held-out device is genuinely unseen. Pooled row-level coverage of nominal 90% intervals, with the median multiplicative half-width in brackets.

Model	split conformal	by device	+ distance
random forest	31% (1.23×)	84% (2.46×)	88% (2.76×)
hist grad. boosting	38% (1.24×)	77% (2.29×)	80% (2.53×)
ridge, log-linear	85% (1.34×)	92% (1.40×)	93% (1.43×)
power law, collisionless	88% (1.34×)	92% (1.39×)	93% (1.42×)
IPB98(y,2)*	90% (1.39×)	90% (1.39×)	90% (1.39×)

*historical reference, not refitted within these folds.

the label-level scheme reported 90% and 89%. The label-level number therefore overstates how well this repair calibrates the booster on a device it has never seen. Both linear models are unaffected within a point or two. The cost is visible in the widths: the trees buy that coverage at roughly 2.3 to 2.8× multiplicative half-width against 1.23× under plain split conformal, where the linear models stay near 1.4×.

Across the size cut with the same device grouping the picture is unchanged in kind. The forest and booster remain far from nominal at 41% and 12% by device and 57% and 21% with distance scaling, better than the label-level 29% and 4% but nowhere near 90%, while the constrained power law holds at 95% and 93%. Changing the calibration unit cannot repair a shift the calibration set does not contain, which is the point of the section.

Among the models refitted within folds, the constrained power law of Sec. 9 is the only one in Table 8 whose plain split-conformal coverage stays near nominal at the ITER-size-matched cut, and its intervals hold there under every scheme. IPB98(y,2) also holds at 88%, but it is a historical reference rather than a fold-fitted competitor.

11 Robustness on rows the standard analysis set excludes

Everything above rests on one file, which is a central limitation of the preceding analysis. The same OSF project publishes the full database revision, DB5.2.3, with 14153 rows against STD5's 6228, and we pin it by its own SHA-256. Matching on (tokamak, shot, time) shows STD5 is an ELMy-H-mode quality selection out of it and leaves two populations this study had not analysed: **5358 H-mode rows STD5 does not contain** (12 labels, zero row overlap) and **3860 ohmic and L-mode rows** in a different confinement regime, which we score against ITER89-P [21] because an H-mode law is the wrong baseline for an L-mode plasma.

This replication has one failure mode that would not announce itself. If the row match against STD5 fails, every STD5 row stays in the "disjoint" arm, that arm becomes a superset of the data Sec. 5 was computed on, and it reproduces Sec. 5 by construction. The matcher is therefore checked positively as well as negatively: STD5 matched against itself must overlap completely, and the disjoint arm must share zero rows.

The ranking inverts in both arms. On the disjoint H-mode rows the best cross-validated model beats the published law by 42%, a margin of the same size as the 36% over that law reported above and computed on rows it never saw, and the best model on an unseen label is IPB98(y,2). On the ohmic and L-mode rows the cross-validation gain over ITER89-P is 67% and the best model on an unseen label is the collisionless power law of Sec. 9. Counting label–model pairs where a tree ensemble loses to the published law on an unseen label: 19 of 24 and 10 of 10.

Row disjointness is not discharge disjointness, and the paper's own argument says the discharge is the unit that matters. Of the 5358 H-mode rows, 425 (7.9%) are slices of one of 336 discharges that STD5 samples at a different time. Dropping those discharges outright leaves 4933 rows across 2864 discharges that share no shot with STD5 at all, and the arm survives: the best cross-validated model still beats the published law, by 39% against 41% on the full arm, and the best model on an unseen label is still IPB98(y,2), with the tree ensembles at 0.526 and 0.552 against the power law's 0.332. The reversal does not depend on the shared discharges.

So the reversal is not an artifact of the standard set's selection criteria, and not a property of ELMy H-mode or of IPB98(y,2) specifically. It is emphatically *not* an independent database: both arms come from the same ITPA collection and the same devices, and the five-machine arm is too small to carry a claim alone.

12 Locked predictions at three device operating points without matching H-mode observations

Everything above is retrospective. To make the disagreement checkable rather than merely described, we record what each model predicts for three real devices, with intervals, a date, and a digest over the input rows so that a later edit leaves a mark. The parameter sets are published design values, and for SPARC and ITER, evaluating IPB98(y,2) on the adopted design vector reproduces the reference confinement time its own source quotes, to 0.6% and 2.9% respectively; the JT-60SA source supplies operating parameters rather than a quoted confinement time. Those are checks that the right machine is being described. None of the three has a *measured* H-mode thermal confinement time at the operating point tabulated here, which is what makes these predictions rather than retrodictions.

How these intervals are built. They are not the plain split-conformal intervals of Sec. 8, and deliberately so: Table 6 measures those covering 3% of rows across the ITER-size-matched cut, so quoting them for a next-step device would publish an interval already known to fail in exactly this situation. Each row instead uses the distance-scaled label-level scheme of Sec. 10, which gives the best size-cut calibration for the models that fail most severely there, applied with the whole database as the training set. Two disclosures belong

Table 10: Predicted energy confinement time, in seconds, with nominal 90% intervals. Locked 31 August 2026 against a fixed dataset digest. Operating points are published design values: SPARC from [3] (V2 primary reference discharge), ITER from [1] ($Q = 10$ inductive baseline), and JT-60SA from its Research Plan [22] (full-current inductive scenario). Intervals are distance-scaled held-out-label calibration intervals, not the plain split-conformal intervals of Table 6, and are pointwise intervals for a new observation; their construction is specified in the text below. The complete input vector for each device is recorded in `results/forecast.json` under the content digest quoted below.

	SPARC 1.85 m	JT-60SA 2.96 m (operating)	ITER 6.2 m
IPB98(y,2)*	0.765 [0.58, 1.00]	0.479 [0.35, 0.66]	**3.591 [2.66, 4.85]**
power law, collisionless	0.724 [0.55, 0.96]	0.428 [0.31, 0.59]	2.837 [2.09, 3.84]
ridge, log-linear	0.720 [0.53, 0.98]	0.444 [0.32, 0.62]	2.858 [2.07, 3.95]
random forest	0.136 [0.03, 0.53]	0.449 [0.23, 0.87]	**0.435 [0.19, 0.98]**
hist grad. boosting	0.305 [0.14, 0.69]	0.418 [0.24, 0.74]	0.444 [0.24, 0.84]

with that. The scheme was chosen after comparing the three in Table 8, so these intervals are exploratory rather than prospectively selected, in the same sense as the constraint rung of Sec. 9. And it calibrates over database labels rather than physical devices, so these are not device-level conformal intervals: Table 9 measures what that distinction costs, and for the tree ensembles it is not negligible. The label-level scheme is retained here because it is the one the forecast was locked under on 31 August 2026, and re-deriving the intervals afterwards would make the lock meaningless. A reader who prefers the device-level calibration should read the tree rows of Table 10 as correspondingly optimistic in width. For a given model, each eligible label k is held out in turn, the model is refitted on the remaining labels, and the absolute log residuals on k are recorded together with each of its rows' Mahalanobis distance d from that inner fit set. A scale $\sigma(d) = \exp(c_0+c_1 d)$ is fitted to those pairs by least squares on $\log|r|$, dropping exactly-zero residuals; the conformity scores are the residuals divided by $\sigma(d)$; and the half-width is the $\lceil(n+1)(1-\alpha)\rceil$-th smallest such score at $\alpha = 0.10$, the same finite-sample rank used in Sec. 8. The interval reported for a device is that quantile multiplied by $\sigma(d)$ evaluated at the device's own distance, applied symmetrically in $\log\tau$ and exponentiated. The model whose point prediction is tabulated is then fitted once on all 6228 rows. For IPB98(y,2) the same construction is used on its analytic residuals, so its interval is comparable to the others rather than merely coexisting with them. Every interval is therefore an empirical pointwise calibration interval for a new observation rather than a finite-sample-valid conformal interval. Three things stand in the way, and only the last is about ITER: the conformity scores remain clustered row-level scores, the distance scale is estimated from the calibration residuals it normalises, and every calibration unit is smaller than ITER along the size direction, which is the limitation Sec. 10 measures directly.

Table 10 illustrates the practical consequence of the disagreement in one place. On JT-60SA, which sits inside the database's size range and is already operating, all five models agree to within 15%. On ITER, 1.82× beyond it, they disagree by a factor of **8.3**.

For the random forest that gap cannot be closed by tuning the standard estimator, and Sec. 5 says why: it predicts by averaging training targets, so its output cannot exceed the largest one, which here is 1.321 s, and no hyperparameter moves that ceiling. The gradient booster carries no corresponding arithmetic bound but produces a similarly low point prediction. Because its prediction cannot exceed the largest training target, 1.321 s, the random forest cannot reach the 2.84 s to 3.59 s range the linear and physics-informed models predict for this ITER operating point, and its nominal 90% interval there runs from 0.19 s to 0.98 s, containing none of those predictions.

The JT-60SA column becomes checkable first, though not against any measurement published as of the 31 August 2026 forecast lock: the first campaign ran L-mode divertor plasmas, and its published power balance reports the energy confinement time following L-mode scaling [23], whereas the row above is an H-mode prediction scored against IPB98(y,2). Checking it needs H-mode operation at a comparable design point, which the research plan [22] places in a later campaign and which the overview of first operation and forward plans [24] discusses in view of ITER and DEMO; we deliberately do not put a date on that here, because the schedule is not ours to assert and the prediction is locked against a parameter vector rather than against a calendar.

Each row of Table 10 carries its own interval and is assessed separately against it, so the range quoted here names one model rather than the set. A stationary JT-60SA H-mode near this design point reporting a thermal confinement time outside 0.349 s to 0.657 s would fall outside the IPB98(y,2) reference row's stated 90% predictive interval. The fitted rows are assessed at their own bounds, which are narrower for the constrained power law (0.313 s to 0.587 s) and wider for the random forest (0.232 s to 0.867 s). A single observation outside a nominal 90% interval is evidence against that forecast rather than a refutation of it: a perfectly calibrated model still places about one observation in ten outside such an interval, so a single miss is inconsistent with the prediction at the stated interval level and no more. Because all five intervals overlap across roughly this range, a measurement inside it separates none of them: this row tests whether any model is right, not which, and that is why the ITER column carries the argument.

Nothing above anticipated one feature of this table: SPARC sits further from the training data than ITER does, at a Mahalanobis distance of 6.9 against 4.7, despite being the smaller machine. Its 12.2 T field is 2.1 times the largest toroidal field anywhere in the cleaned STD5 rows, 5.82 T on Alcator C-Mod, and every other label in the database tops out at or below 4.80 T. Size is not the only direction in which a next-step device leaves the data behind, and an extrapolation audit organised around size alone would have missed it.

13 Flexibility, or long-range saturation?

Section 5.1 left two readings open, and the polynomial ladder there supports the flexibility one only for a single kind of flexibility: polynomial degree under an isotropic penalty, swept over nine penalty strengths $10^{-3}, 10^{-2}, \ldots, 10^{5}$ at each degree without rescue; the full grid is tabulated in `results/flexibility_sweep.csv`. Every model scored to this point confounds two properties. Flexibility is how much structure a model can learn beyond a power law. *Boundedness* is what it does far outside the training data. We call a model *saturating* when its predictions stop tracking the features once the input leaves the training support, either because they cannot leave a fixed range or because they revert to a fixed value. The random forest is the strict case, provably confined to $[\min y_{\text{train}}, \max y_{\text{train}}]$; the gradient booster carries no such theorem but never left that range on any split scored here (Sec. 5.1); an RBF process reverts to its prior mean, a third mechanism with the same consequence. What the three share is the absence of a trend that continues outside the data, and that is what this section tests. The tree ensembles here are flexible and saturating; ridge is neither; a degree-3 polynomial is flexible and non-saturating but divergent, which is why its worst label is 4.601. No model to this point was flexible, non-saturating and well behaved at long range at once.

A Gaussian process (GP) [25] supplies one, and the property is a choice of kernel rather than a tuning parameter. We fit three over the same log features, with the same optimizer on the same rows. The intended difference between them is whether the kernel admits a trend that continues at long range. Adding a dot-product term also enlarges the function class, so this is not a clean single-factor experiment and we do not read it as one. A radial basis function (RBF) kernel decays to zero with distance, so the posterior returns to its prior mean, which centring places at the training mean: bounded. A dot-product kernel is Bayesian linear regression in the log features, which is a power law: unbounded and inflexible. Their sum is both.

Specification. The three kernels are

$$\begin{aligned} k_{\mathrm{RBF}}(x,x') &= \sigma_r^2 \, \exp\Bigl(-\tfrac{1}{2}\|x-x'\|^2/\ell^2\Bigr) + \sigma_n^2\delta_{xx'}, \\ k_{\mathrm{lin}}(x,x') &= \sigma_l^2 \, (x^\top x' + \sigma_0^2) + \sigma_n^2\delta_{xx'}, \\ k_{\mathrm{lin+RBF}}(x,x') &= k_{\mathrm{lin}} + k_{\mathrm{RBF}} - \sigma_n^2\delta_{xx'}, \end{aligned} \tag{5}$$

with a single isotropic length scale ℓ shared across the standardised log features, so the RBF component is isotropic rather than automatic-relevance-determining. Features are standardised inside the pipeline, so each fold's scaler is fitted on that fold's training rows alone and ℓ is in those units. The prior mean is constant and `normalize_y` is on, which puts it at the training mean of $\log\tau$: this is what makes "reverts to the prior" and "returns the average training target" the same statement, and the parallel with a tree ensemble exact. Observation noise enters as an additive `WhiteKernel` rather than through a fixed jitter. Hyperparameters are fitted by maximising the log marginal likelihood with scikit-learn's default L-BFGS-B optimizer, `n_restarts_optimizer` $= 0$, from the shared initialisation $\sigma_l^2 = 1$, $\sigma_0 = 1$, $\sigma_r^2 = 0.5$, $\ell = 1$, $\sigma_n^2 = 0.05$, under the bounds $\sigma_l^2, \sigma_0 \in [10^{-3}, 10^3]$, $\sigma_r^2 \in [10^{-3}, 10^3]$, $\ell \in [10^{-2}, 10^3]$ and $\sigma_n^2 \in [10^{-6}, 1]$. Every rung starts from the same values for every term it has, so the three fits differ only in which terms exist. The optimisation runs on a subsample of exactly 1000 training rows drawn without replacement by `numpy.random.default_rng(42)`, or on the whole fold when it holds fewer; the kernel is then frozen and the exact GP is conditioned on *every* training row of that fold, which is one Cholesky factorisation. No held-out row enters either stage. Hand-chosen hyperparameters are not adequate here, since a length scale an order of magnitude from the one the data supports makes the third rung look like a failure. Intervals quoted for these models are equal-tailed posterior intervals on $\log\tau$ that include the fitted observation-noise term, exponentiated to τ.

The control behaves as intended: the linear-kernel process and ridge agree to 0.0000 under cross-validation and on a held-out label, and to 0.0007 at the ITER-size-matched cut, which is what licenses reading the other two rows as consequences of the kernel. The bounded rung then fails exactly as a tree does. It beats the power law under cross-validation and scores 1.948 at the ITER-size-matched cut, worse than predicting a constant, with error tracking distance at $\rho = +0.65$. The mechanism is the one in Sec. 5: a random forest cannot exceed the largest training target, and an RBF process reverts to its prior mean. Both are structural long-range behaviours rather than tuning shortfalls, and on the held-out rows this one's predictions carry half the spread of the truth.

Adding a dot-product term to that same kernel keeps the nonlinear RBF component it already had and supplies an unbounded linear trend alongside it. Flexibility is retained

model	CV	held-out	ITER cut	ρ
GP, linear + RBF	**0.112**	0.218	**0.191**	−0.01
GP, RBF only	0.142	0.541	1.948	+0.65
GP, linear only	0.181	0.214	0.278	−0.06
random forest	0.128	0.465	0.938	+0.85
IPB98(y,2), historical ref.	0.199	0.188	0.194	−0.49
mean baseline	0.869	0.994	1.459	+0.56

Table 11: log-RMSE under the three splits, and the rank correlation of per-label error against extrapolation distance. The intended difference across the first three rows is whether the kernel admits a trend that continues at long range; adding the linear term also enlarges the function class. The three span a factor of ten at the ITER-size-matched cut. IPB98(y,2) is a historical reference fitted to the DB2.8 predecessor of this database [2] and is not refitted within these folds.

$m(x)$ in Eq. (6)	CV	held-out	ITER cut
constant	0.142	0.541	1.948
fitted power law	**0.115**	**0.212**	**0.278**

Table 12: One RBF covariance family and fitting procedure, two mean functions. The constant-mean row reproduces the RBF rung of Table 11 to six decimals, which is the control: the decomposition is the same model rewritten. Changing only the mean-function specification moves the ITER-size-matched cut by a factor of 7.0.

rather than held exactly fixed, since a linear kernel enlarges the function class as well as changing the asymptote; what the comparison isolates is the availability of a continuing trend. It gives the best cross-validated score in this work, 0.112 against the random forest's 0.128; it scores 0.191 at the ITER-size-matched cut, better than the analytic law, which is a historical reference fitted to the DB2.8 predecessor and not refitted within these folds, and second only to the constrained fit of Sec. 9 at 0.183; and its error is flat against distance at $\rho = -0.01$.

13.1 The same experiment with the nonlinear component specified identically

Adding a dot-product term enlarges the function class, so the ladder above suggests the mechanism without isolating it. One rearrangement removes that objection. Write the model as a mean function plus a residual process,

$$\log \tau = m(x) + g_{\mathrm{RBF}}(x), \tag{6}$$

and specify g identically in both arms: same RBF covariance family, same tuning subsample, same seed, same optimizer. Only m changes, from a constant to a fitted log-linear power law. The two fitted processes are not the same function, because each is conditioned on its own mean's residuals and each re-optimises its hyperparameters on those. What is held fixed is the nonlinear capacity available to the model and the procedure that fits it, so no part of the difference below can come from one arm having more of either.

Table 12 reports the two arms. Take the control first: with a constant mean, Eq. (6) returns 0.141533, 0.540967 and 1.947535, which is the RBF row of Table 11 to every digit computed. Nothing has changed except how the model is written. Giving the same process a power-law mean then moves the ITER-size-matched cut from 1.948 to 0.278, a factor of 7.0, and improves the cross-validated score as well, 0.115 against 0.142. Because the residual GP has the same covariance family, the same capacity and the same fitting procedure in both rows, the difference cannot be attributed to adding nonlinear model capacity.

The second row's cut score is worth reading closely: 0.277823 is plain ridge's score at that cut, to six decimals. Far outside the data the RBF residual has reverted to zero, which is what an RBF process does, and the model falls back to its power-law mean, which is the mechanism of this section stated as a single number. A saturating model reverts to something; the failure in Table 11 is that a constant mean makes that something a constant, and the repair is not to make the model less flexible but to give it something worth reverting to.

Across the model families tested here, this resolves the choice Sec. 5.1 left open: extrapolation failure tracks long-range saturation rather than flexibility alone. The set of models in Sec. 5 is measured correctly, and every flexible member of it was either bounded or divergent, so nothing in it could have separated the two properties; that is why the reading there was flagged as provisional rather than asserted. "Do not use a flexible model out of distribution" is the wrong lesson. "Do not use a model whose predictions stop tracking the features once the input leaves the training data" is the right one, and the random forest's training-target ceiling is its strictest case. This does not displace the constrained power law, which still wins the ITER-size-matched cut with no within-surface hyperparameter to tune, and the process does not beat the power law on a held-out label (0.218 against 0.214). Figure 4 summarises the three kernels.

A Gaussian process also supplies an interval by construction rather than by calibration, so Sec. 8's question can be put to a model that was never calibrated. At the ITER-size-matched cut, at nominal 90%, the flexible unbounded rung covers 92.5%, where split-conformal intervals give the random forest 3% and the gradient booster none. It is the only flexible model tested whose *native* posterior intervals reach nominal coverage out of distribution without conformal recalibration. That is a statement about where the interval comes from rather than about coverage alone: the constrained power law of Sec. 9 also holds near nominal at this cut (91%, Table 8), but through an externally calibrated split-conformal interval rather than one the model supplies itself. The bounded rung is the instructive failure: it is the one model that does become vague out of distribution, with a half-width four times any other, and it still covers 16.7%, having reverted to a prior mean nowhere near the answer. Width was never the problem.

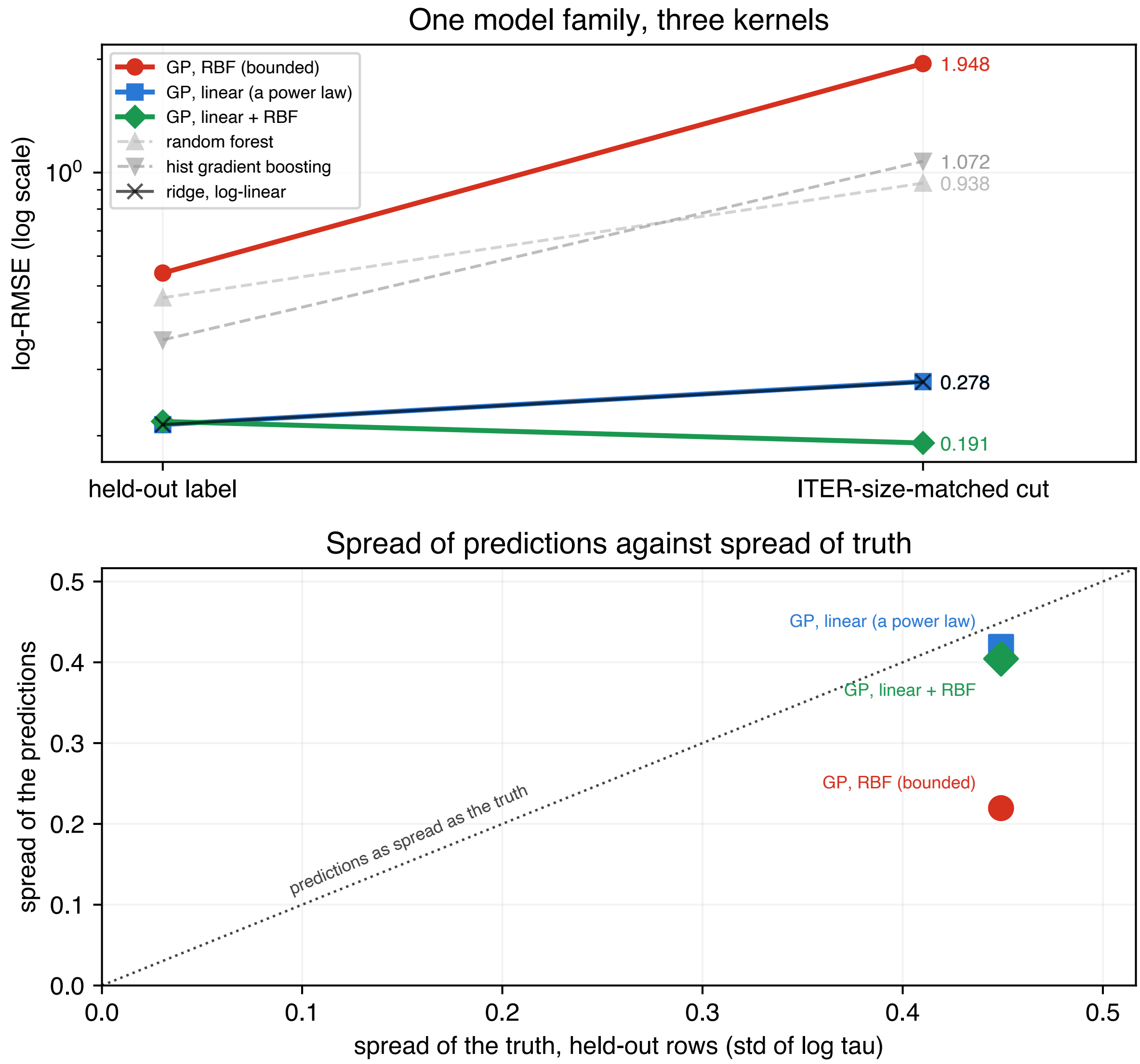

Figure 4: One model family, three kernels. Gaussian processes with a radial-basis-function (RBF) kernel, a linear (dot-product) kernel, and their sum. Top: log-RMSE on a held-out label and at the ITER-size-matched cut. The kernels differ in whether they permit a trend that continues at long range; adding the linear component also enlarges the function class. Bottom: spread of the predictions against spread of the truth on the held-out rows. A model whose posterior has reverted to its prior mean predicts nearly the same value everywhere, so its spread sits near zero however good its in-range fit was: the same failure a tree ensemble shows when it cannot exceed its largest training target, reached by a different route.

14 Discussion

The practical consequence of Table 1 is narrow and specific. It is not that machine learning has nothing to offer confinement scaling, and not that the models scored here are badly built. It is that conventional within-database evaluations do not measure the quantity the law is used for. Cross-validation grouped by discharge, or by shot, or by time slice, holds out rows from machines that remain in the training set. It measures interpolation within known devices. A scaling law is used to size a device that does not exist, which is extrapolation to an unknown one, and on this database the two comparisons do not merely differ in magnitude: they order the candidate models in opposite directions. A gain reported under the first is not evidence about the second.

That suggests a reporting convention rather than a modelling one. Any confinement model offered as an alternative to a power law can be scored under leave-one-label-out on the same rows and the same features, at a cost of one refit per unit, and the two numbers can be printed together. Where they agree, the interpolation number is informative. Where they disagree, as here by a factor of 3.6, the disagreement is the result. Two diagnostics that need no held-out device are worth reporting alongside: the distance of the target operating point from the training distribution, which tracked per-label error at $\rho = +0.85$ for the forest and not at all for the power law, and whether the model's prediction can leave the range of its training targets at all, which for a random forest it provably cannot.

The positive findings point the same way, and both are about the far field rather than about capacity. Constraining the exponents to a Connor–Taylor surface and adding a linear component to a Gaussian process kernel are unrelated operations, and both supply long-range structure. They are not equivalent in what else they change: the constraint removes freedom from the fit, while the linear kernel also enlarges the GP's function class, so that rung on its own cannot separate the asymptote from the capacity. The mean-function ablation of Sec. 13.1 is the cleaner test, since there the nonlinear covariance family and the procedure fitting it are held fixed and only the long-range mean changes. It should still be read as a statement about the families tested, not about flexible models in general.

For practice, the cheaper option was also the stronger one out of distribution. A power law refitted on the same rows is a stronger baseline out of distribution than any tree ensemble scored here, and it costs a single least-squares solve; the estimator hardly matters, since ordinary least squares on the independent columns reproduces the reported ridge to within 0.0012 on every split. Where a flexible model is wanted, the evidence here favours one that keeps trending, and the linear-plus-RBF Gaussian process was both the best cross-validated model in this study, at 0.112, and the only flexible model whose native posterior intervals reached nominal coverage out of distribution without conformal recalibration; the constrained power law reaches comparable coverage there through split conformal instead. Where a physics constraint is available, imposing it costs 0.001 in in-sample log-RMSE and was the best-scoring option at the size cut, on a constraint rung chosen after those scores had been seen.

Nothing in the argument so far rules out reading all of it as a fact about tokamaks, so the same audit was run on two scaling laws from biology, reported in full in the Supplementary Material. On Kleiber's law for mammalian metabolic rate against body mass [26], with taxonomic order in the part played by the tokamak, both tree ensembles lose to both power laws at all 8 mass cuts and the training-range bound binds at every one; the ranking inversion does not reproduce, because with a single predictor the trees never win the interpolation split and so have no margin to give up. A predictor-count ladder on the Biomass And Allometry Database for woody plants [27], holding 3599 plants fixed and varying only how many of four measurements the models see, tests that explanation directly: the interpolation margin crosses zero between two and three predictors, the inversion appears at exactly that crossing and at every rung above it, and the power law wins the held-out species at all four rungs regardless [28].

None of this is settled by one database. In that nested predictor ladder the inversion appears exactly where the flexible model first gains an interpolation advantage, and not before. The experiment does not establish predictor count itself as the causal variable or identify a universal threshold, since each added column carries biological information as well as a dimension. What it does support is the weaker and more useful reading: where a flexible model has no interpolation advantage to lose, the extrapolation hazard is present with no warning sign at all. The extrapolation failure itself reproduced everywhere it was tested; it is the inversion, not the failure, that requires a condition to appear.

15 Limitations

The refit population is not IPB98's, and the refit exponents should not be read as a correction to it. Two machines (JET and ASDEX Upgrade with their variants) supply 77% of rows. Only 13 of the 18 database labels are scored under leave-one-label-out; the five excluded are the smallest and most unlike the rest, so the reported gap is if anything optimistic. Thirteen labels is a small bootstrap and the paired differences are the more reliable statistic. The training-target bound of Sec. 5.1 is a theorem for the random forest and an observation for the gradient booster: boosting sums tree outputs rather than averaging targets, so nothing confines it to the training range, and what is reported here is only that it stayed inside that range on all 14 splits scored. The ITER-size-matched cut reproduces ITER's size ratio and nothing else: ITER is a burning plasma dominated by alpha self-heating at a gain no row here approaches, so clearing this bar is necessary and not sufficient. The per-label coverages in Sec. 8 are proportions over as few as 39 rows and carry binomial uncertainty of ten points or more; the pooled contrasts are far larger than that noise, the within-column ordering is not. The constraint hierarchy of Sec. 9 has an optimum in its middle that nothing here predicted and that cross-validation cannot select, so the result is that a constraint helps rather than that this constraint is the right one. The robustness arms of Sec. 11 are disjoint in rows but not independent in provenance: both come from the same ITPA collection and the same devices, and the five-machine arm is too small to carry a claim alone. The H-mode arm is disjoint in rows and not, as first constructed, in discharges; Sec. 11 reports the rerun with all 336 shared discharges removed, which the reversal survives. The locked forecast of Sec. 12 is a record, not a validation; only JT-60SA can be checked against measurement in the near term, that check requires an H-mode campaign rather than the completed L-mode one and we put no date on when that arrives, and it is the one device of the three that sits inside the database's size range. The two biological replications carry their own limitations, stated in full in the Supplementary Material: the metabolic-rate arm has one predictor and eleven groups that are phylogenetically related rather than independent, which the comparative-biology literature handles with phylogenetically controlled regression [29] and we do not, and the plant ladder is one ordering of one selectively intersected set of predictors, so it establishes that a crossing exists and that the inversion tracks it, not that three is a threshold. Neither establishes a rate for anything. Every power-law fit here treats the engineering parameters as measured without error, which the confinement-scaling literature has long objected to because these predictors carry real measurement uncertainty. Refitting

Eq. (1) by orthogonal distance regression, which spreads the error across the predictors instead, moves the exponents enormously: up to 5.6 on the inverse aspect ratio, with the major radius going from 1.58 to 5.40. It also more than doubles the in-sample error, 0.454 against 0.181. Both are symptoms of one thing rather than a better fit. With no per-variable measurement uncertainties published for this database, an errors-in-variables (EIV) estimator has nothing pinning it down along the weak singular direction of Sec. 4, which carries 0.3% of the design's variance, and it runs away along exactly that direction. The honest statement is that these are least-squares exponents, that an EIV fit would differ, and that this database does not supply what computing a credible one would need. Every fit also weights rows equally, so JET and ASDEX Upgrade set most of what each model learns even though the target is transfer to a device neither resembles; refitting the power law with each training label weighted equally does not help, moving its leave-one-label-out score from 0.214 to 0.259, so the row imbalance is not an easy source of its advantage, though that is one weighting of one model and does not settle the question. The Gaussian-process intervals of Sec. 13 are equal-tailed posterior intervals on $\log \tau$ that include the fitted observation-noise term and are exponentiated to τ; that is the closest available analogue of a conformal interval, which is also a statement about a new observation rather than a latent mean, but the two are not the same construction and the comparison should be read as such. Those kernels are tuned by marginal likelihood on a seeded subsample of each training fold, with only the hyperparameters coming from the subsample and the posterior then conditioned on the whole fold, which is stable across the range we could afford (under 4% in every hyperparameter between 750 and 1500 rows) but is not a proof that the full-data optimum agrees; and "linear plus RBF" is one reading of "a power law with a bounded correction" among several, with an anisotropic kernel, a Matern kernel and a linear part constrained as in Sec. 9 all untested. The last of these is the obvious next experiment, since the constraint and the kernel are the two things that work here and nothing has tried them together.

16 Conclusion

On real confinement data, the model that wins cross-validation by 29% over a log-linear power law refitted inside the same folds loses to that same power law on all 13 held-out database labels and on all 11 physical devices once the JET and ASDEX Upgrade variants are collapsed, and to the published IPB98(y,2) law as well, though that law is a historical reference fitted to the DB2.8 predecessor of this database and was not refitted within these folds. Three separate diagnostics point at that failure and none of them is a tuning fix; the mechanism they converge on is long-range saturation. At the size extrapolation that separates this database from ITER, the two tree ensembles that lead the conventional cross-validation comparison land closer to a constant predictor than to the power law, and their nominal 90% intervals cover 3% and 0% of the held-out rows at unchanged width. In these tests the power-law form survives leaving the data behind despite fitting the observed rows less accurately under conventional cross-validation. A bounded correction on its residuals improves it along the size-extrapolation direction tested here, and the bounded range that disqualifies trees as predictors is what limits how much extrapolation error that correction can contribute.

The repair that scored best, though, is not additional model flexibility at all: it is a dimensional constraint on the existing power law. Constraining the exponents to the surface implied by the collisionless Connor–Taylor similarity assumptions gave the lowest observed error of any fit tested here at the ITER-size-matched cut, cost 0.001 in sample, and has no within-surface hyperparameter to tune. That rung was selected after the held-out scores at this cut had been seen, as Sec. 9 states, so it is the lowest error among the surfaces examined rather than a prediction that this surface is the right one. Supplying the same physics as the Gaussian prior tested here is worth substantially less: the constraint restricts the fit to a surface those assumptions permit where the prior only favours a location within parameter space, and here the surface transfers better. The inversion itself survives several separate robustness checks: rows the standard set does not contain, a second confinement regime, collapsing the 13 database labels into 11 physical devices, and either aggregation of the score. The inversion is therefore robust to the validation, aggregation and row-selection choices tested here, rather than an artifact of a single bookkeeping choice. What the models disagree about is recorded rather than argued: on a machine inside the database's size range they agree to within 15%, and on ITER they differ by a factor of 8.3.

Two results temper the diagnosis and neither weakens the practical advice. In the nested predictor ladder, the extrapolation failure persists at every predictor count tested while the ranking inversion does not: the inversion needs the flexible model to win the interpolation split first, so a field whose scaling law has one predictor faces the same danger with no warning at all. And separating flexibility from long-range saturation shows the failure belongs to the second. A bounded RBF process fails as the trees do, while adding an extrapolative linear component to that process gives the best cross-validated score here, holds up at the ITER-size-matched cut, and is the only flexible model in this work whose native posterior intervals reach nominal coverage out of distribution without conformal recalibration. That comparison enlarges the function class as well as changing the asymptote, so the cleaner test is the ablation that fixes the nonlinear component's specification and varies only the mean function. It moves the ITER-size-matched cut by a factor of 7.0, and the better arm lands exactly on the power law's score there, because far from the data its correction has reverted to zero and left the trend behind. The lesson is not that flexible models cannot be trusted beyond their data. It is that a model whose predictions stop tracking the features once the input leaves the training data is structurally unsuited to a problem where the target itself has to leave that range. The two are easy to confuse because the flexible models that win the conventional interpolation comparison here are the saturating ones, while the non-saturating polynomial control becomes unstable in extrapolation instead.

Acknowledgments

This work uses the ITPA Global H-mode Confinement Database, assembled and maintained by the ITPA Confinement Database and Modelling Topical Group and by the teams operating the contributing tokamaks. The author thanks them, and the authors of Ref. [2], for releasing the standard analysis set publicly: an independent recomputation of this kind is possible only because they did. The replications outside fusion use the BAAD compilation [27], released under CC0, and the metabolic-rate compilation deposited on Figshare that the Supplementary Material describes. Neither the ITPA group nor any contributing team has reviewed or endorsed this analysis, and its conclusions are the author's alone.

Use of generative AI. Two generative AI systems were used in preparing this work, and each is named with its version and its role.

Anthropic Claude Opus 5 (model `claude-opus-5`) assisted with developing, debugging and executing author-specified analysis code in the accompanying repository, and with drafting and editing passages of this manuscript.

OpenAI GPT-5.6 Sol was used to critique complete drafts against journal policy and for scientific and historical accuracy, to support the literature review, and to identify methodological issues and propose robustness checks. Several of its suggestions were adopted after the author judged them worth pursuing. They include the corrected account of how IPB98(y,2) was fitted, in Sec. 5 and Sec. 9; the recomputation of the distance correlation over the 11 physical devices in Sec. 5.1; the device-level calibration of Sec. 10; and several of the references in Sec. 4 and Sec. 12, which the author then located and checked against their published records.

Neither system is an author. The author designed the study, decided which suggestions to pursue, specified and implemented every analysis reported here, verified every reported number against the generated artifacts, checked every reference against its published record, and takes full responsibility for the content, the conclusions and any error in them. No text was incorporated without the author reading and verifying it.

Funding

This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests

The author declares no competing interests.

Author contributions

Sole author. L.S. designed the study, wrote the analysis code, performed the analysis, and wrote the manuscript.

Data availability

Four third-party datasets underlie this work. None is redistributed here; each is fetched from its public source by a script in the repository and verified against a pinned SHA-256 on load, so every number above refers to one specific set of bytes rather than to whatever a host currently serves.

- **ITPA Global H-mode Confinement Database, standard analysis set STD5 (DB5.2.3).** The basis of every fusion result. Openly available from the OSF deposit *International Global H-Mode Confinement Database* [30], described in Ref. [2]. File `hdb5_std5.csv`.
- **The full DB5.2.3 revision.** Used only for the replication of Sec. 11, which scores rows STD5 excludes. It is a separate download from the same OSF project (`https://osf.io/download/zhwa3/`) and is not a superset this paper generated. File `hdb5_db523.csv`.

- **Mammalian basal metabolic rate.** The Figshare deposit of Capellini, Venditti and Barton [31]. Used in the Supplementary Material. File `allometry_bmr.txt`.
- **Biomass And Allometry Database (BAAD) v1.0.1.** Released CC0; the data paper is Ref. [27] and the archive analysed here is release v1.0.1 [32]. Used in the Supplementary Material. File `baad_data.zip`.

The pinned fingerprints are:

file	SHA-256
`hdb5_std5.csv`	`67601c2da5c51f90cf6298ff499cccc7` `4d09ac80c2b98c7dde0d8db3ebb9ac5b`
`hdb5_db523.csv`	`7a48e34379663e3e298924990f05cd8f` `16b8581516bcfa7bb8f438f83ae80ab6`
`allometry_bmr.txt`	`ab22d9ae8f35e96d7fbf5d8e53053d64` `7f0594d74f8917888169993ce668571e`
`baad_data.zip`	`0375f012475658c039e3dcef398951e8` `f68cdeb5430eec8cd65548965496cfe2`

The locked forecast of Sec. 12 additionally carries a content digest over its input rows, so a later edit to the operating points leaves a mark:

```
960d537ee3df8780afd2054e51d01a268
f8803aca64a830dba37fd1ff5319192
```

All code, the full writeup with every table and limitation, and the scripts that regenerate every number and figure in this paper and its Supplementary Material are openly available under the MIT licence at `https://github.com/lsalcedo100/FusionFlux`, and the exact version used for this paper is archived with a persistent identifier [33]. The analysis code is pinned as firmly as its inputs: every number and figure above was produced at commit

```
11049ff19abe4002462661687dd24bd15eb5acfb
```

Each analysis script also writes the dataset fingerprint it verified into its own JSON output under `results/`, so a regenerated number carries the provenance of the bytes it was computed from.

References

[1] ITER Physics Expert Group on Confinement and Transport, ITER Physics Expert Group on Confinement Modelling and Database and ITER Physics Basis Editors, "Chapter 2: Plasma confinement and transport," *Nucl. Fusion* **39**, 2175 (1999). doi:10.1088/0029-5515/39/12/302

[2] G. Verdoolaege *et al.*, "The updated ITPA global H-mode confinement database: description and analysis," *Nucl. Fusion* **61**, 076006 (2021). doi:10.1088/1741-4326/abdb91

[3] A.J. Creely *et al.*, "Overview of the SPARC tokamak," *J. Plasma Phys.* **86**, 865860502 (2020). doi:10.1017/S0022377820001257

[4] J. Kates-Harbeck, A. Svyatkovskiy and W. Tang, "Predicting disruptive instabilities in controlled fusion plasmas through deep learning," *Nature* **568**, 526 (2019). doi:10.1038/s41586-019-1116-4

[5] J.X. Zhu *et al.*, "Hybrid deep-learning architecture for general disruption prediction across multiple tokamaks," *Nucl. Fusion* **61**, 026007 (2021). doi:10.1088/1741-4326/abc664

[6] J. Degrave *et al.*, "Magnetic control of tokamak plasmas through deep reinforcement learning," *Nature* **602**, 414 (2022). doi:10.1038/s41586-021-04301-9

[7] A. Murari, M. Peluso, M. Gelfusa and I. Lupelli, "Symbolic regression via genetic programming for data driven derivation of confinement scaling laws without any assumption on their mathematical form," *Plasma Phys. Control. Fusion* **57**, 014008 (2015). doi:10.1088/0741-3335/57/1/014008

[8] Z. Wang *et al.*, "Power-law-anchored residual learning for H-mode energy confinement time in tokamaks: interpolation and parameter-defined extrapolation," arXiv:2608.20684 (2026). doi:10.48550/arXiv.2608.20684

[9] H. Nam and J. Seo, "Machine learning based energy confinement time extrapolation via multi-tokamak database," *Fusion Eng. Des.* **221**, 115369 (2025). doi:10.1016/j.fusengdes.2025.115369

[10] E. Gao *et al.*, "Bayesian neural networks for predicting tokamak energy confinement time with uncertainty quantification," *Nucl. Fusion* **65**, 084001 (2025). doi:10.1088/1741-4326/ade8fd

[11] X. Xiong, K. Zhang, Y. Gao, L. Zhang, X. Chen and S. Liu, "Bayesian optimization based machine learning for predicting tokamak energy confinement time with quantification of feature importance," *Fusion Eng. Des.* **227**, 115740 (2026). doi:10.1016/j.fusengdes.2026.115740

[12] D.R. Roberts *et al.*, "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography* **40**, 913 (2017). doi:10.1111/ecog.02881

[13] L. Breiman, "Random forests," *Machine Learning* **45**, 5 (2001). doi:10.1023/A:1010933404324

[14] J. Hall, P. Zhang and G. Verdoolaege, "Confinement scaling with machine size in the updated ITPA global H-mode energy confinement database," in *Proc. 49th EPS Conf. on Plasma Physics*, Bordeaux, Europhys. Conf. Abstr. **47A**, P1.024 (2023).

[15] J. Hall, P. Zhang, Y. Zhang, C. Angioni and G. Verdoolaege, "Confinement scaling with machine size in the updated ITPA global H-mode energy confinement database," poster P2.1097, *52nd EPS Conf. on Plasma Physics*, Edinburgh (2026).

[16] V. Vovk, A. Gammerman and G. Shafer, *Algorithmic Learning in a Random World*, Springer (2005). doi:10.1007/b106715

[17] R.J. Tibshirani, R. Foygel Barber, E.J. Candès and A. Ramdas, "Conformal prediction under covariate shift," *Adv. Neural Inf. Process. Syst.* **32**, 2526 (2019).

[18] Y. Ovadia *et al.*, "Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift," *Adv. Neural Inf. Process. Syst.* **32**, 13969 (2019).

[19] J.W. Connor and J.B. Taylor, "Scaling laws for plasma confinement," *Nucl. Fusion* **17**, 1047 (1977). doi:10.1088/0029-5515/17/5/015

[20] B.B. Kadomtsev, "Tokamaks and dimensional analysis," *Sov. J. Plasma Phys.* **1**, 295 (1975).

[21] P.N. Yushmanov *et al.*, "Scalings for tokamak energy confinement," *Nucl. Fusion* **30**, 1999 (1990). ITER89-P. doi:10.1088/0029-5515/30/10/001

[22] JT-60SA Research Unit, *JT-60SA Research Plan: Research Objectives and Strategy*, Version 5.0 (2021). `https://www.jt60sa.org/wp/wp-content/uploads/2021/02/JT-60SA_Res_Plan-5.pdf`

[23] M. Yoshida *et al.*, "Overview of plasma operation and control studies in the first plasma campaign of JT-60SA," *Plasma Phys. Control. Fusion* **67**, 065010 (2025). doi:10.1088/1361-6587/add563

[24] J. Garcia *et al.*, "Overview of first JT-60SA plasma operation and plans in view of ITER and DEMO," *Nucl. Fusion* **66**, 116011 (2026). doi:10.1088/1741-4326/ae74e1

[25] C.E. Rasmussen and C.K.I. Williams, *Gaussian Processes for Machine Learning*, MIT Press (2006). doi:10.7551/mitpress/3206.001.0001

[26] M. Kleiber, "Body size and metabolism," *Hilgardia* **6**, 315 (1932). doi:10.3733/hilg.v06n11p315

[27] D.S. Falster *et al.*, "BAAD: a Biomass And Allometry Database for woody plants,"

Ecology **96**, 1445 (2015). Released CC0. doi:10.1890/14-1889.1
[28] G.B. West, J.H. Brown and B.J. Enquist, "A general model for the structure and allometry of plant vascular systems," *Nature* **400**, 664 (1999). doi:10.1038/23251
[29] J. Felsenstein, "Phylogenies and the comparative method," *Am. Nat.* **125**, 1 (1985). doi:10.1086/284325
[30] ITPA Confinement Database and Modelling Topical Group, *International Global H-Mode Confinement Database*, version 5.2.3 [dataset], Open Science Framework (2021). `https://osf.io/drwcq`
[31] I. Capellini, C. Venditti and R.A. Barton, "Supplement 1: data on basal metabolic rate with experimental animal body mass" [dataset], Figshare (2016). doi:10.6084/m9.figshare.3549807.v1
[32] D.S. Falster *et al.*, *BAAD: a Biomass And Allometry Database for woody plants* [dataset], Figshare (2015). doi:10.6084/m9.figshare.c.3307692. Release v1.0.1, `https://github.com/dfalster/baad/releases/tag/v1.0.1`
[33] L. Salcedo, *FusionFlux: the model that wins on cross-validation loses on a new machine*, version v0.4.0, Zenodo (2026). doi:10.5281/zenodo.22545928